

Introduction to Autonomous Vehicles

Lecture 1 – Thursday September 28, 2016

Objectives

When you have finished this lecture you should be able to:

- Understand the main characteristics of autonomous systems.
- Familiarize with mobile robot as a platform to study autonomous systems.
- Recognize basic terminology, classes and applications of mobile robots.
- Recognize challenging problems of mobile robotics.

Outline

- Autonomous Systems
- From Manipulators to Mobile Robots
- Mobile Robot Classifications
- Challenging Problems of Mobile Robots
- Summary

Outline

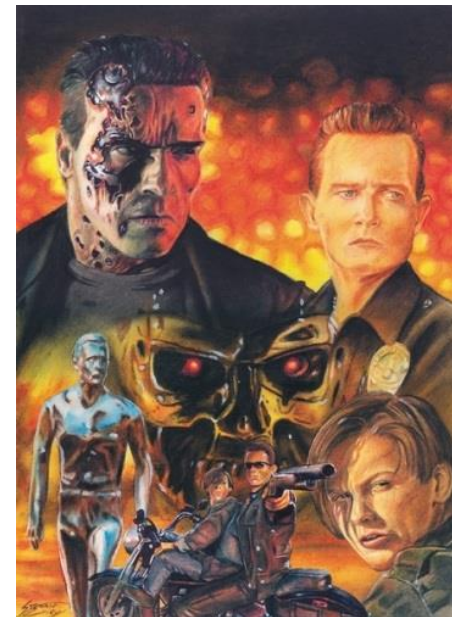
- **Autonomous Systems**
- From Manipulators to Mobile Robots
- Mobile Robot Classifications
- Challenging Problems of Mobile Robots
- Summary

Autonomous Systems

- **Autonomy** (Ancient Greek: auto- “self” + nomos- “law”) is a desirable feature in modern systems.

Autonomy means that a system can adapt to changes in its environment (the lights get turned off) or itself (a part breaks) and continue to reach its goal.

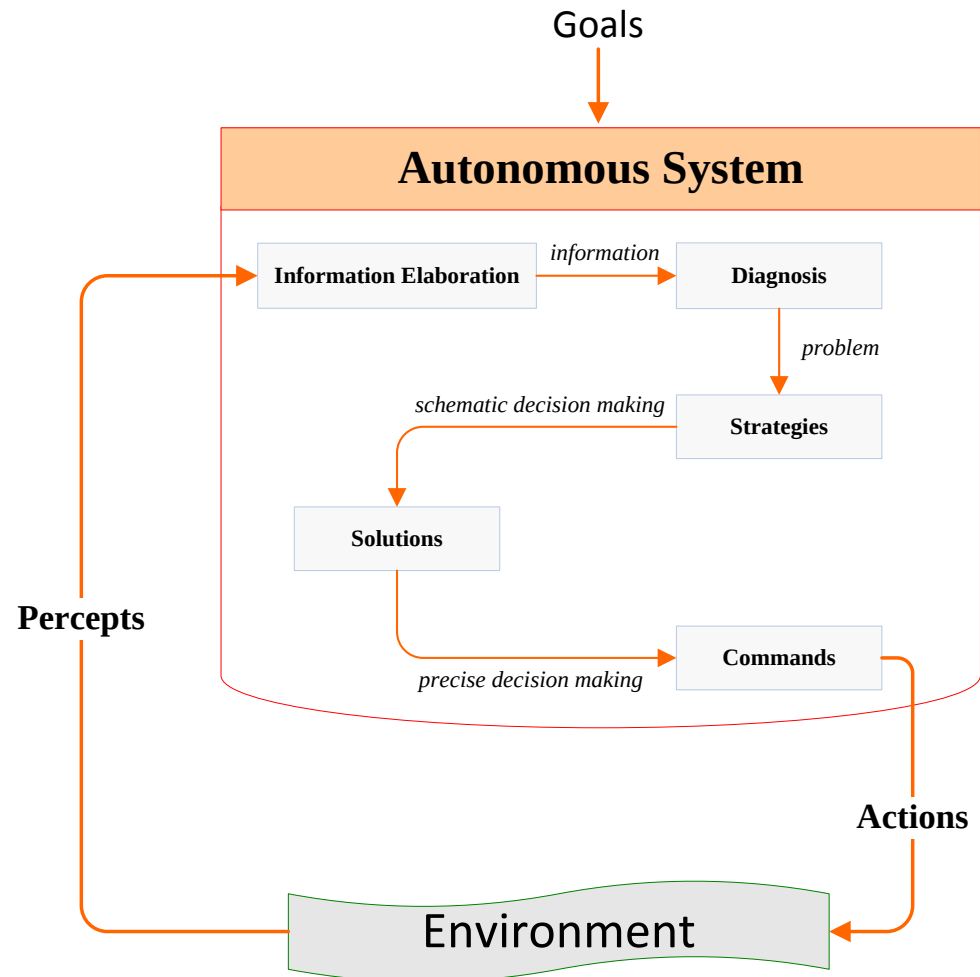
Example: Terminator is an intelligent mechanical creature which can function autonomously . Even after losing one camera (eye) and having all external coverings (skin, flesh) burned off, it continued to pursue its target (Sarah Connor).



Autonomous Systems

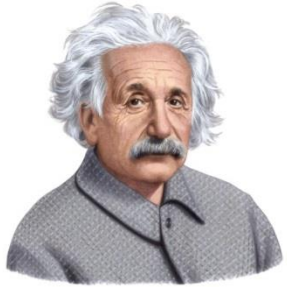
Autonomous systems are **intelligent systems** that can perform desired tasks in **unstructured environments** without continuous explicit human guidance.

These systems have ***know-how*** to solve problems autonomously.



Autonomous Systems

- **Human vs. Machine Intelligence**



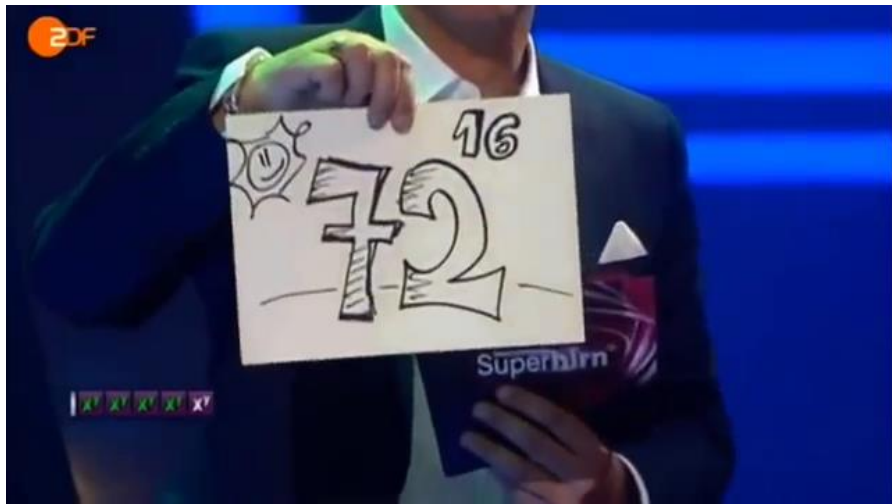
The true sign of intelligence is not knowledge but imagination.

Albert Einstein (1879-1955)



Intelligence is the ability to adapt to change.

Stephen Hawking (1942-)



Autonomous Systems

• Human vs. Machine Intelligence



Brain

Better in

- distinguishing faces
- identifying objects and
- recognizing language sounds

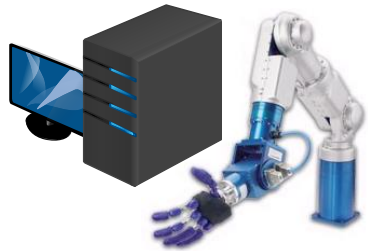


Hamburger



Autonomous Systems

• Human vs. Machine Intelligence



Better in

Machine

- dealing with more complex patterns such as that exist in financial, scientific, or product data.
- operations that require fast, precise, highly repeatable actions
- Working in harsh environments

Robots find applications in “4D tasks”

- Dull
- Dangerous
- Dirty
- Dumb



Autonomous Systems

• Brain Functions

There is more we do NOT know about the brain, than what we do know about the brain

Low-level Functions

- Fully understood
- Localized



Sight



Hearing



Smell



Taste



Touch



High-level Functions

- Partially understood
- Not fully localized



• Perception



• Situation awareness



• Natural Language Understanding



• Pattern Discovery



• Reasoning



• Decision Making



• Planning

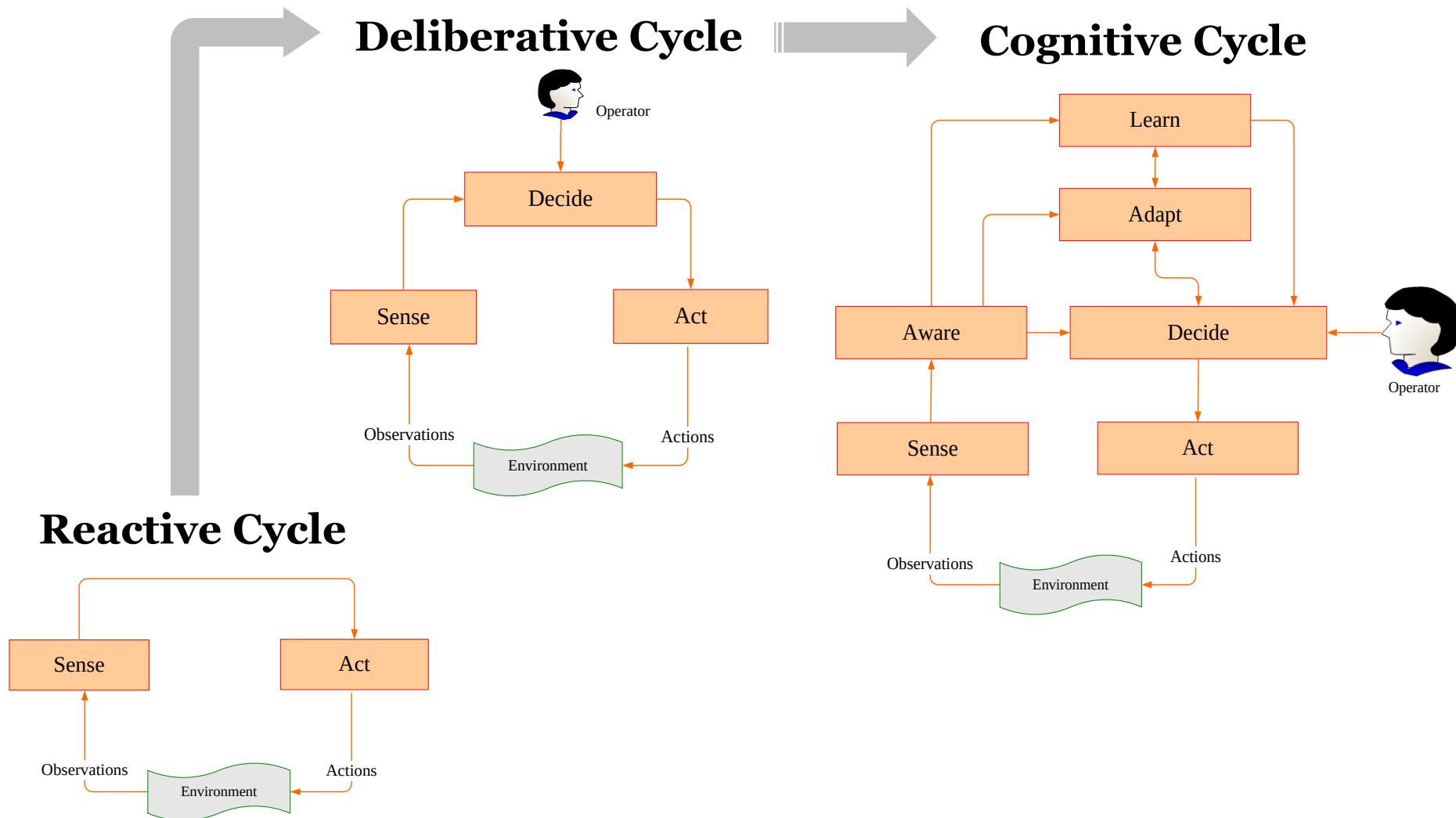


• Learning

• etc.

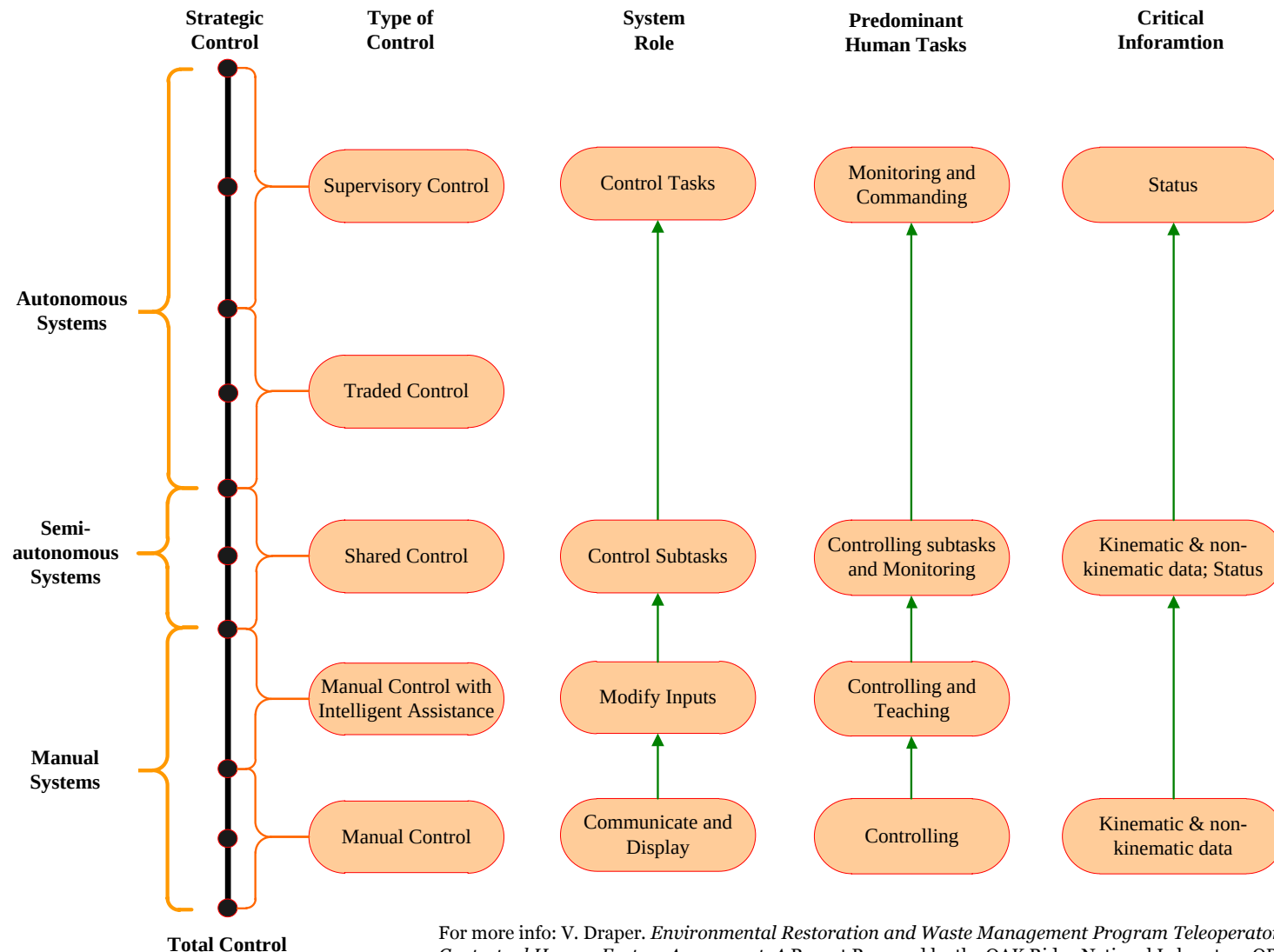
Autonomous Systems

- Levels of Intelligence



Autonomous Systems

• Levels of Autonomy



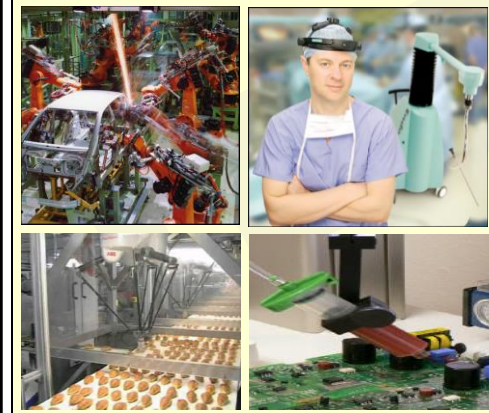
For more info: V. Draper. *Environmental Restoration and Waste Management Program Teleoperator Hand Controllers: Contextual Human Factors Assessment*. A Report Prepared by the OAK Ridge National Laboratory ORNL/TM-12762, May, 1994.

Autonomous Systems

• Levels of Autonomy

- Automotive
- Chemical
- Electronics
- Food

Industry



Field



- Aerial
- Space
- Underwater
- Search and rescue

Service



- Domestic
- Edutainment
- Rehabilitation
- Medical

Source: Bruno Siciliano. Robots Moving Closer to Humans. IEEE Distinguished Ambassador Seminar. Available at: <http://ras-egypt.org/activities.html>

Autonomous Systems

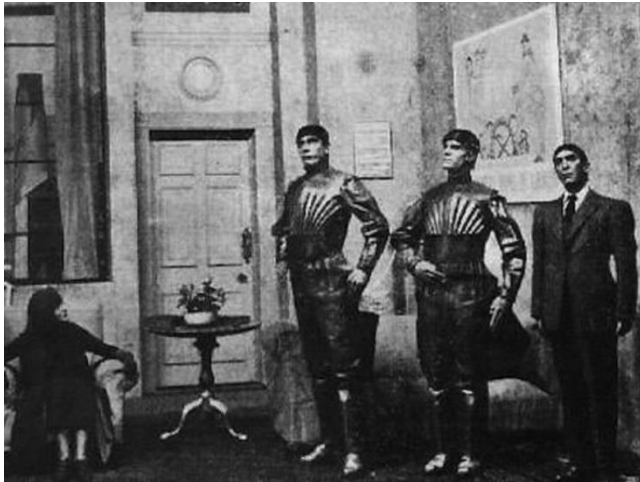
- **Levels of Autonomy**

Autonomy would be a scary feature in military



CAMPAIGN TO **STOP**
KILLER ROBOTS

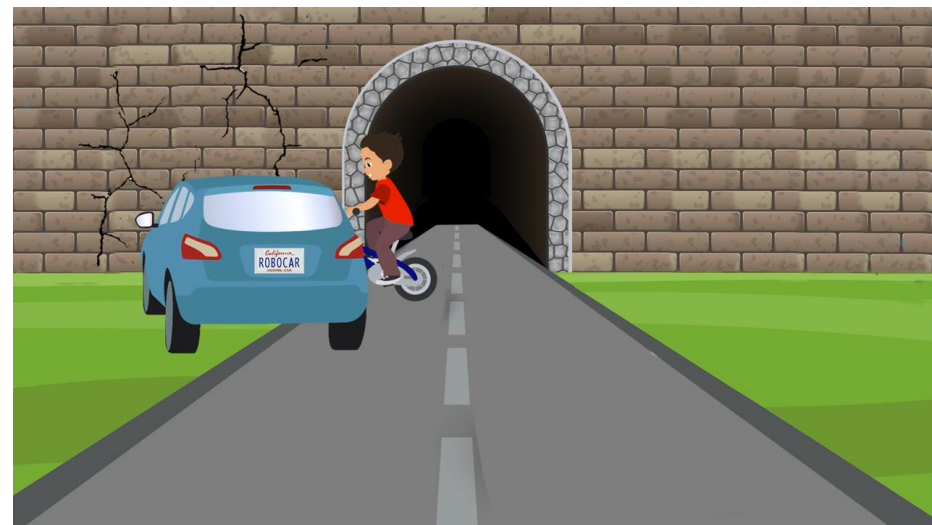
<https://www.stopkillerrobots.org/>



R.U.R. (1920)



Autonomy would be problematic
in driverless car as well
(tunnel/trolley problem)



Autonomous Systems

- **Fully Autonomous systems** must have **self-X capabilities** to make decisions about their conditions, processes and course of actions undertaken to achieve their goals.
- **Self-X capabilities** include:
 - ◇ Self-configuration and management
 - ◇ Self-adaptation or Adaptability
 - ◇ Self-protection, self-diagnosis and self-healing and repair
 - ◇ Self-optimization
 - ◇ Self-synchronization
 - ◇ Self-organization
 - ◇ Self-learning

Autonomous Systems

- **Autonomous vs. Automated Systems**

According to NASA's Robotics, Tele-Robotics and Autonomous Systems Roadmap working group:

*An “**autonomous system**” is defined as a system that resolves choices on its own. The goals the system is trying to accomplish are provided by another entity; thus, the system is autonomous from the entity on whose behalf the goals are being achieved. The decision-making processes may in fact be simple, but the choices are made locally.*

*In contrast, an “**automated system**” follows a script, albeit a potentially quite sophisticated one; if it encounters an unplanned-for situation, it stops and waits for human help, e.g. “phones home”. The choices have either been made already, and encoded in some way, or will be made externally to the system.*

Autonomous Systems

- **Examples: Automated but not Autonomous Systems**



Food and Beverage



Entertainment



Car Fueling



Banking

Autonomous Systems

- **Examples: Autonomous Systems**



Roomba
Autonomous Vacuum Cleaner



Predator Drone
Semi-Autonomous UAV



Plustech
Semi-Autonomous Walking Robot



Predator Drone
Semi-Autonomous UGV



MARS Curiosity
Semi-Autonomous Robot



Fruit-picking Robot
Semi-Autonomous Robot

Autonomous Systems

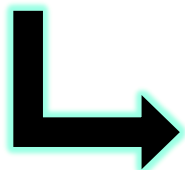
- **Examples: Autonomous Systems**

Q: What is the common ground between these autonomous/semi-autonomous systems?



A: ◇ Goal-directed systems

- ◇ Working in unstructured and dynamic environments
- ◇ Ability to perceive the environment and physically manipulate it autonomously or semi-autonomously
- ◇ Mobility (a key issue)



Intelligent Mobile robots can be used as a platform to study autonomous systems.

Autonomous Systems



Roomba: Autonomous Vacuum Cleaner



Honda ASIMO: Humanoid Robot



Autonomous vehicles are able to **imitate the actions and sometimes the appearance** of an intelligent creature, usually a human. They can **perceive** unstructured and dynamic environments and **act** autonomously.

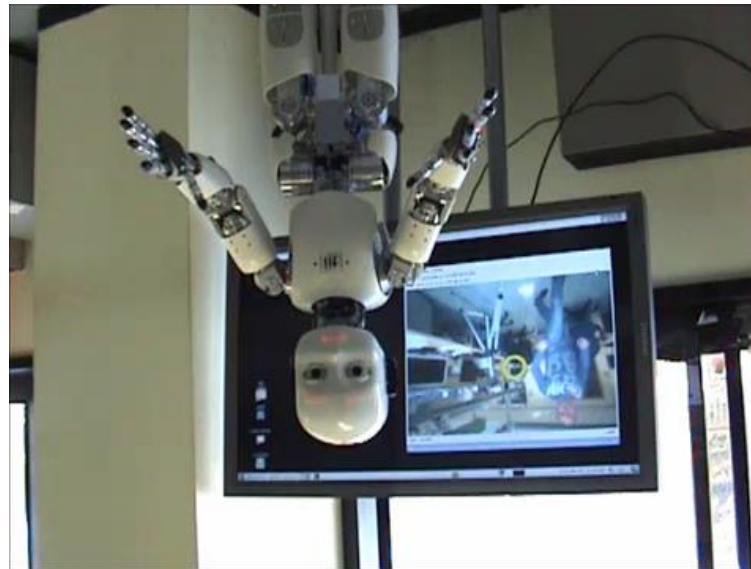
Autonomous Systems

- **Current Robot Skills: Be realistic and don't exaggerate!**

object recognition
capabilities of a
2-year-old child



social understanding of an
8-year-old child



language capabilities of a
4-year-old child



manual dexterity of a
6-year-old child

Source: Bruno Siciliano. Robots Moving Closer to Humans. IEEE Distinguished Ambassador Seminar.
Available at: <http://ras-egypt.org/activities.html>

Outline

- Autonomous Systems
- **From Manipulators to Mobile Robots**
- Mobile Robot Classifications
- Challenging Problems of Mobile Robots
- Summary

From Manipulators to Mobile Robots

A mobile robot is a robot that can move in the real world and can be completely autonomous.

The main features:

- Reprogrammability
- The ability to navigate (Mobility)
- Autonomy

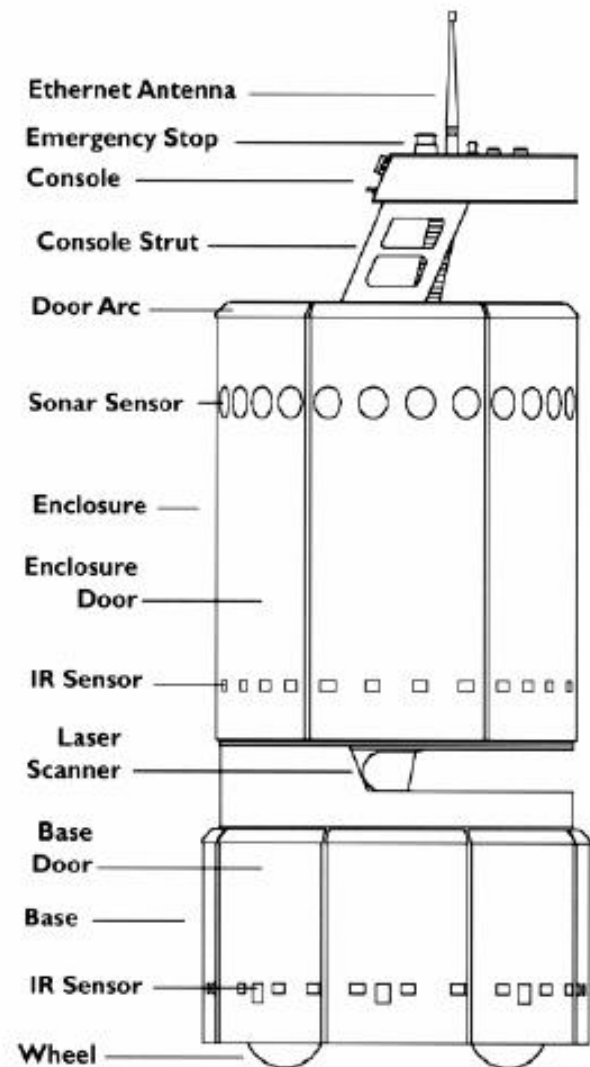


From Manipulators to Mobile Robots

- **Example:** B21 of Real World Interface (now iRobot) is a sophisticated mobile robot with up to three Intel Pentium processors on board. It has all different kinds of on board sensors for high performance navigation tasks.



B21r



From Manipulators to Mobile Robots

Before a general-purpose mobile robot can move freely in a factory, home, farm, or military environment, we have to achieve an **“intelligent” connection of perception** (sensing and understanding the environment) **to action** (planning and control of robot motion within the environment).

P. J. McKerrow. *Introduction to Robotics*. Addison-Wesley, 1992.

From Manipulators to Mobile Robots

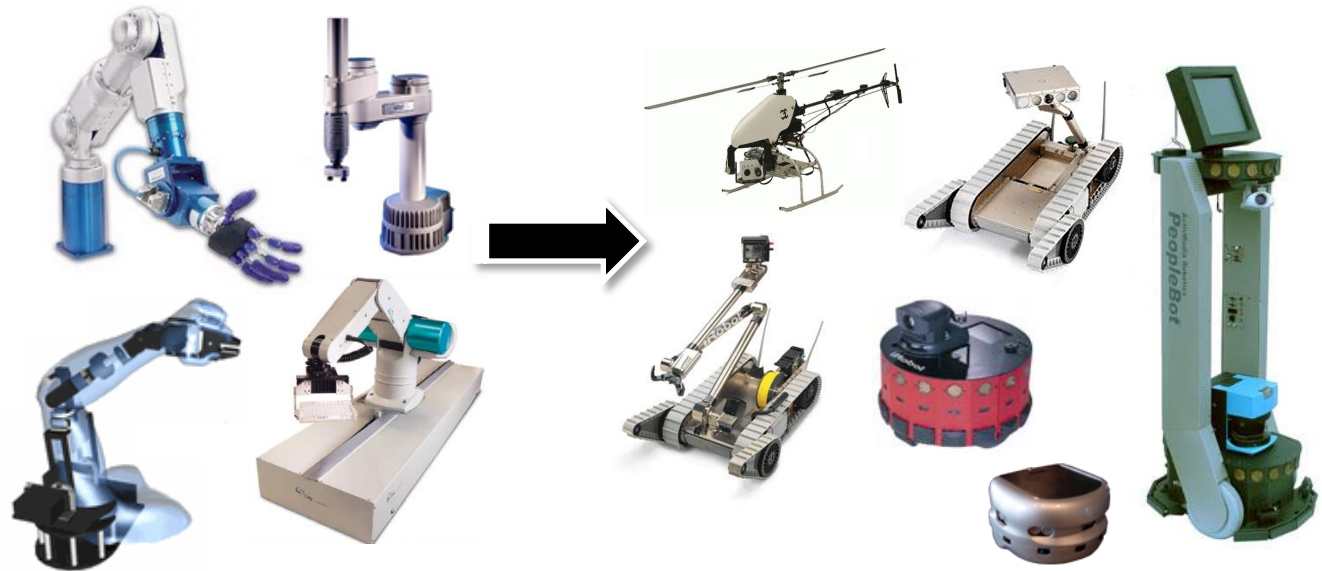
• Intelligent Robot

An intelligent robot is a mechanical creature which can function autonomously.

- ◇ **“Intelligent”**: implies that the robot does not do things in a mindless, repetitive way; it is the opposite of the connotation from factory automation.
- ◇ **“Mechanical Creature”**: means that our scientific technology uses mechanical building blocks, not biological components. It also emphasizes that a robot is not the same as a computer.
- ◇ **“Function Autonomously”**: indicates that the robot can operate, self-contained, under all reasonable conditions without requiring recourse to a human operator.

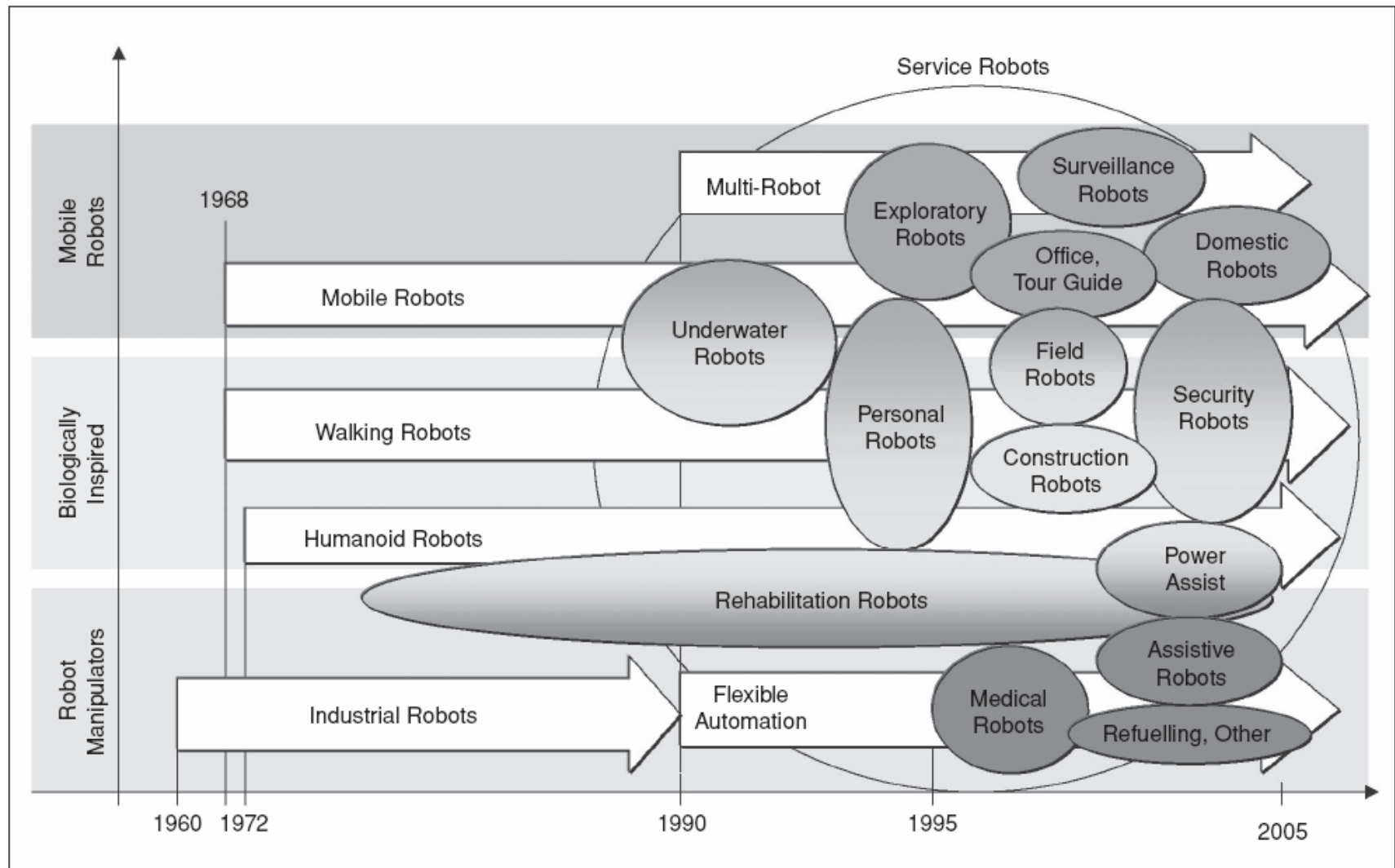
Source: Robin Murphy . *Introduction to AI Robotics*. MIT Press, 2000.

From Manipulators to Mobile Robots



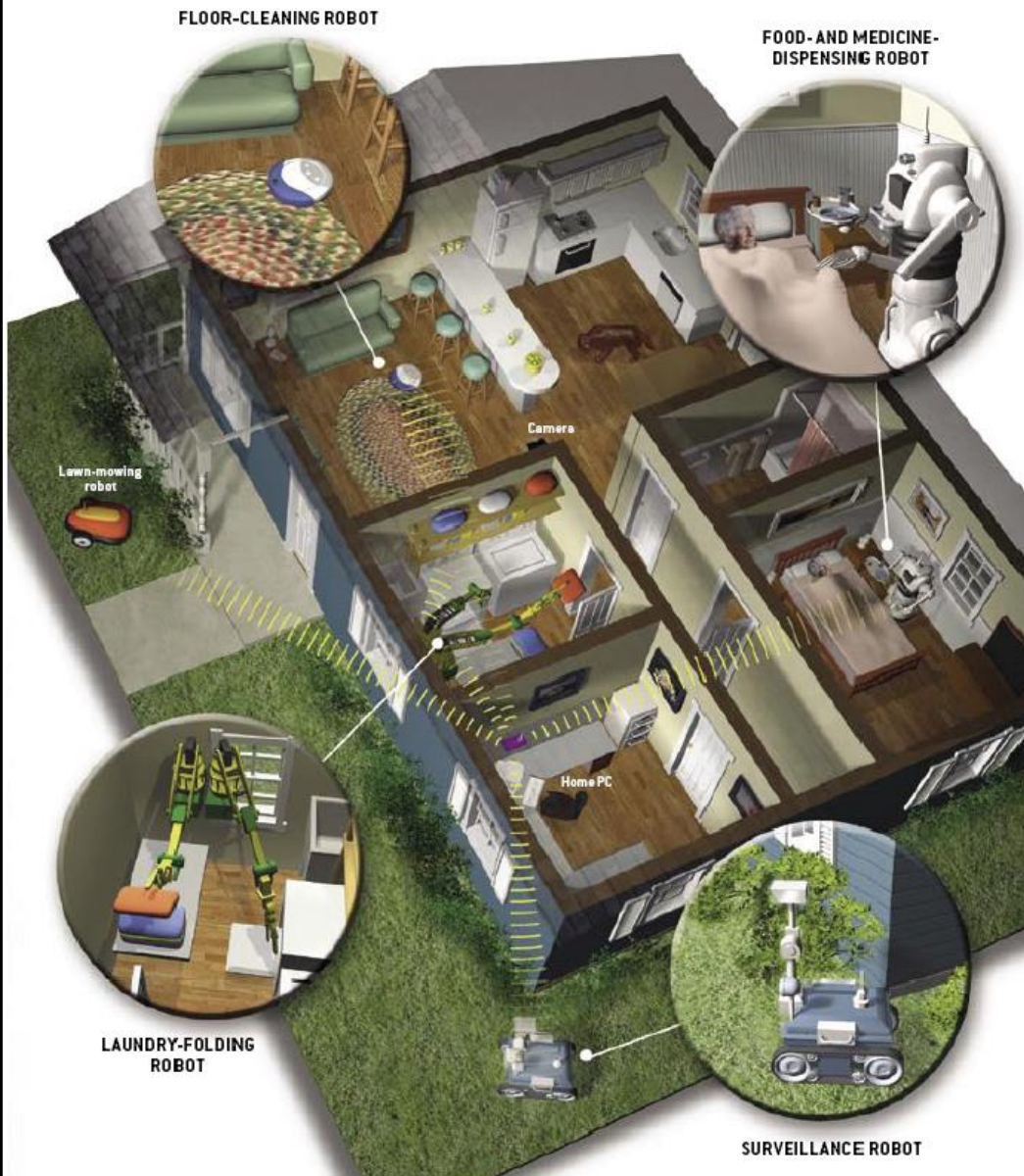
Aspect	Industrial Manipulators	Mobile Robots
Environment	Used to be highly structured and static	Used to be highly unstructured and dynamic
Mobility	Lack or limited	High
Human interaction	Used to be uni-modal and rigid	Used to be multimodal and adaptive
Field maturity	High	Low
Market penetration	High	Low
Application field	Industry	Wide field

From Manipulators to Mobile Robots



Source: ELENA GARCIA, MARIA ANTONIA JIMENEZ, PABLO GONZALEZ DE SANTOS, AND MANUEL ARMADA. "Evolution of Robotics Research: From Industrial Robotics to Field and Service Robotics", IEEE Robotics & Automation Magazine, MARCH 2007.

From Manipulators to Mobile Robots



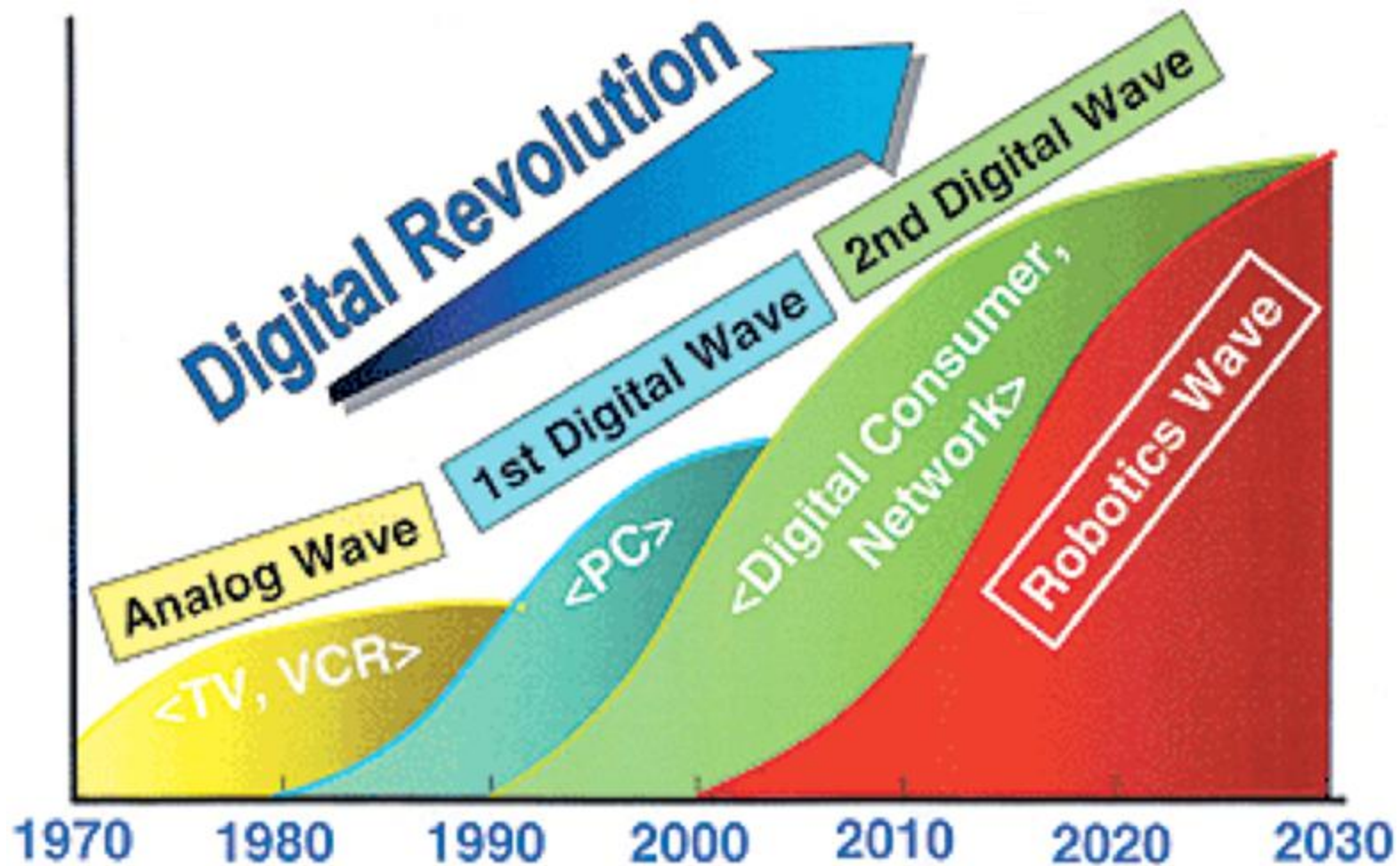
- **A robot in every home**

“As I look at the trends that are now starting to converge, I can envision a future in which robotic devices will become a nearly ubiquitous part of our day-to-day lives. The challenges facing the robotics industry are similar to those we tackled in computing three decades ago.”

Bill Gates, 2007
Scientific American



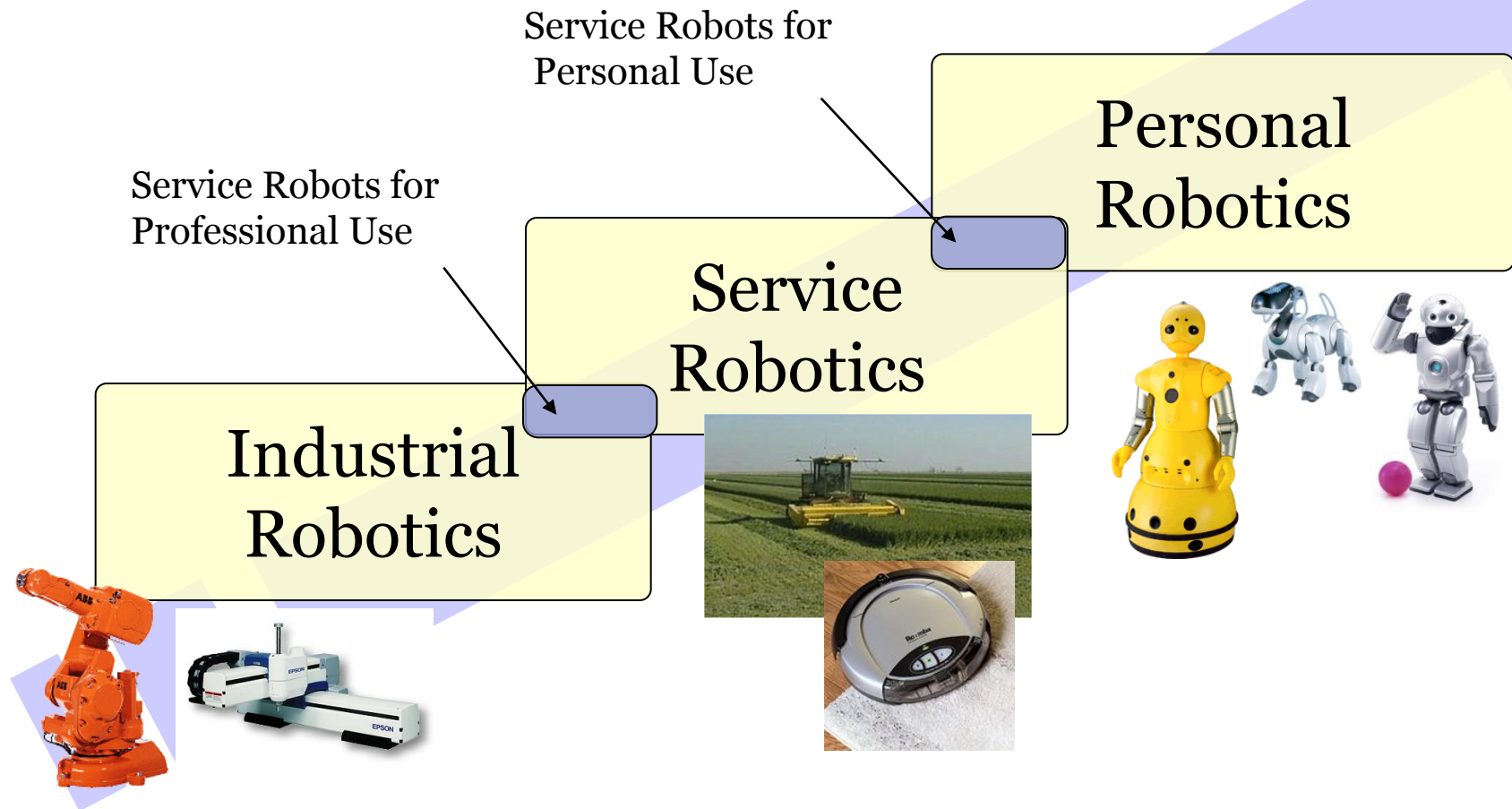
From Manipulators to Mobile Robots



Source: Bruno Siciliano. Robots Moving Closer to Humans. IEEE Distinguished Ambassador Seminar.
Available at: <http://ras-egypt.org/activities.html>

From Manipulators to Mobile Robots

The Evolutionary Stages



From Manipulators to Mobile Robots

- **Service Robots**

Service robots assist human beings, typically by performing a job that is dirty, dull, distant or dangerous.



Service Robots

Service Robots for Professional Use

(Construction robots, demolition, medical, space robots, underwater exploration, cleaning, demining, inspection, fire fighting,...)

Service Robots for Personal Use

(domestic, personal assistant, entertainment robots, vacuum cleaners, etc.)

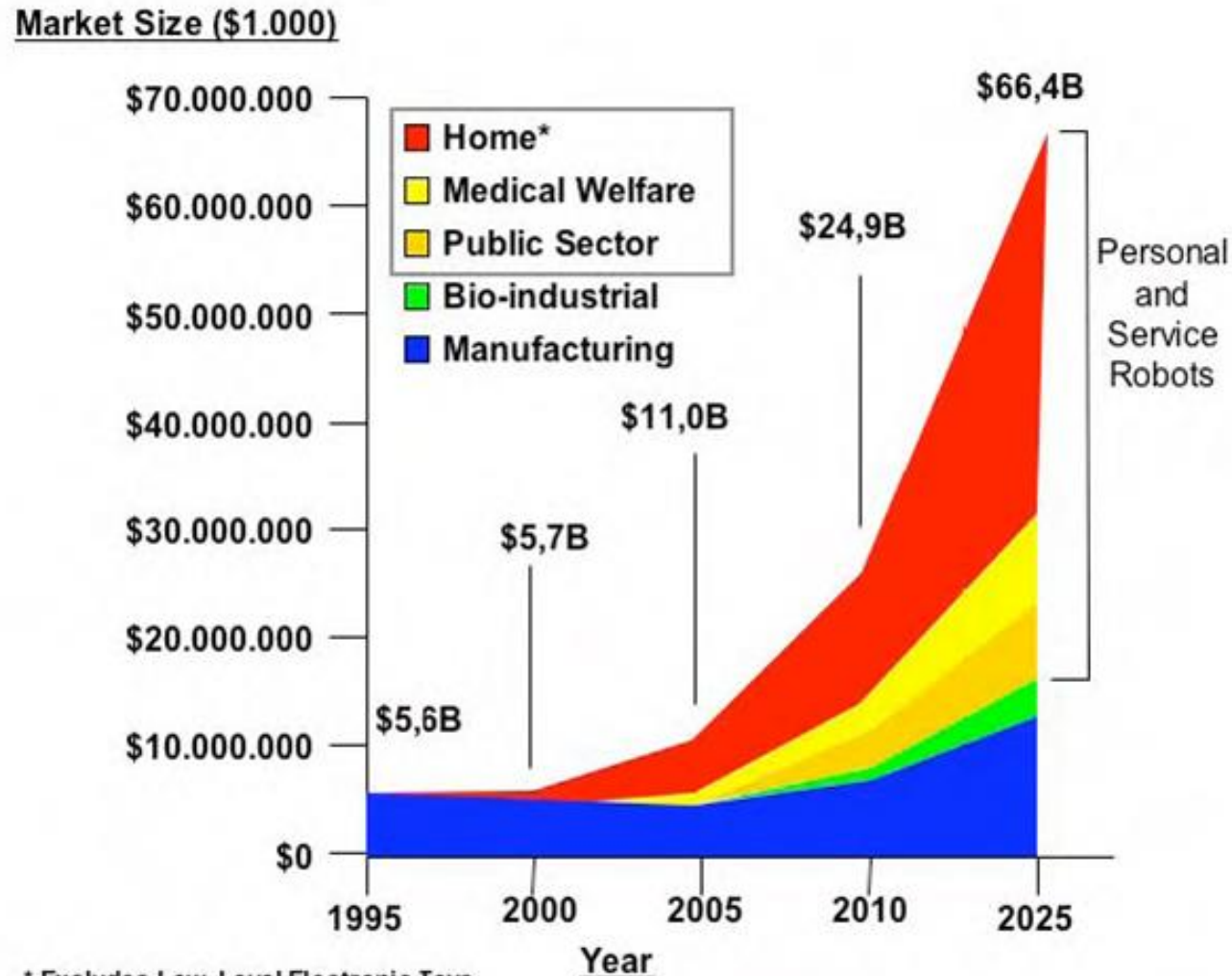


From Manipulators to Mobile Robots

- Recent Statistics

Service robotics is planned to be a major driver of economic growth in the 21st century.

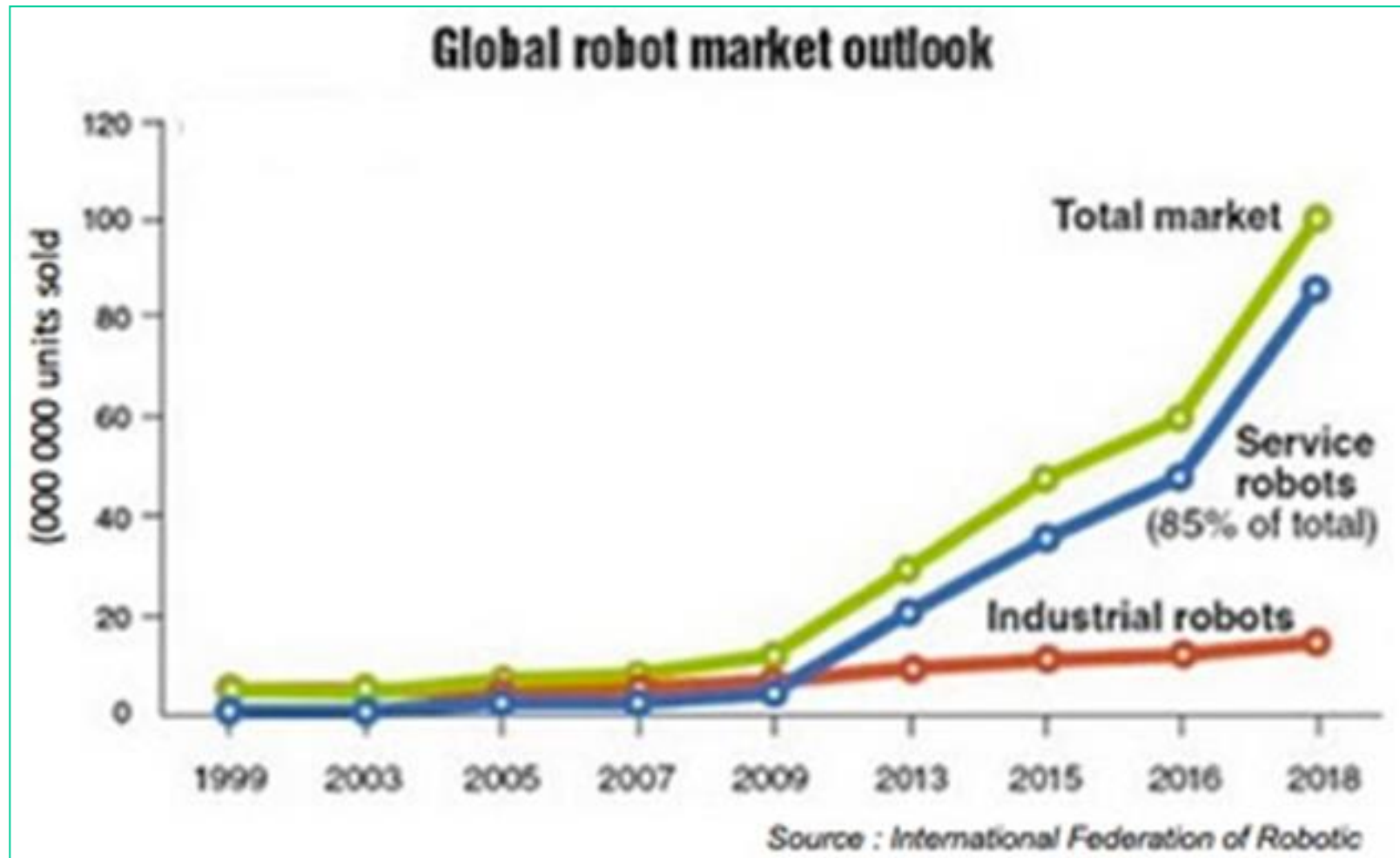
It offers high potential growth, revenue and employment perspectives.



Source : Japan Robotics Association

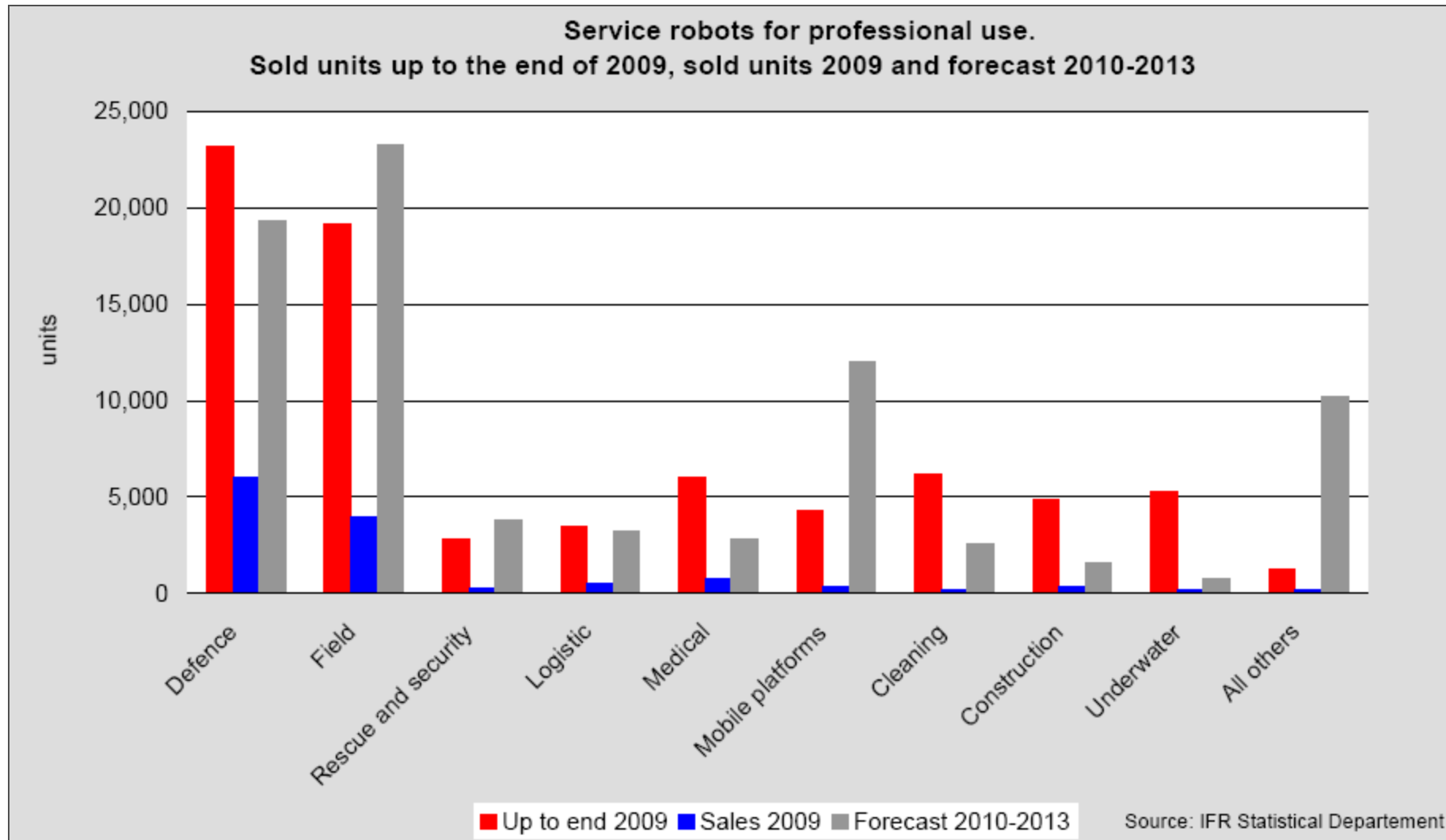
From Manipulators to Mobile Robots

- Recent Statistics



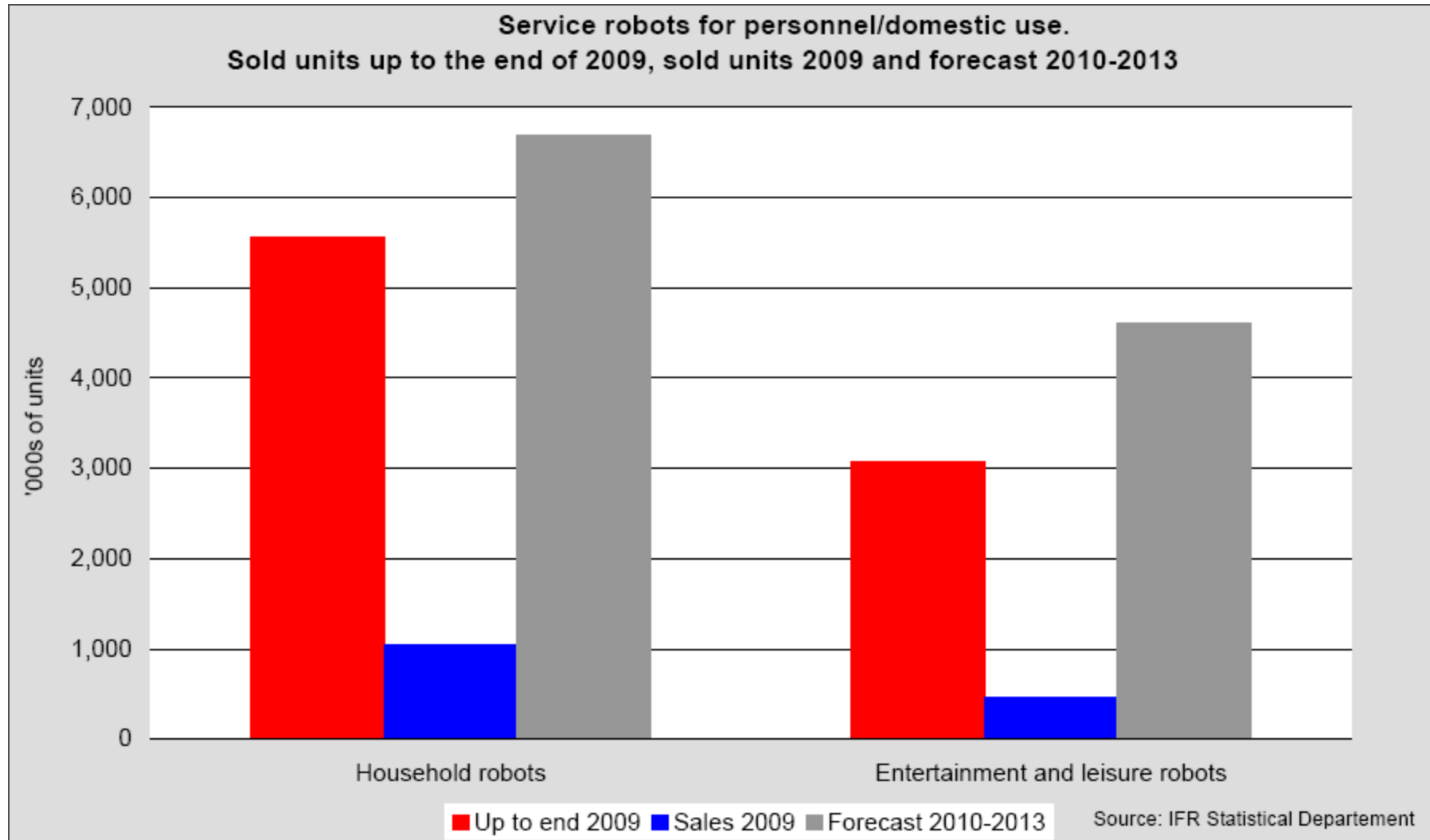
From Manipulators to Mobile Robots

- Recent Statistics



From Manipulators to Mobile Robots

- Recent Statistics



From Manipulators to Mobile Robots

- **Recent Statistics**

For more statistics:

- ROBO Global: <http://roboglobal.com/>
- ROBO Global ETFs: <http://roboglobaletfs.com/>

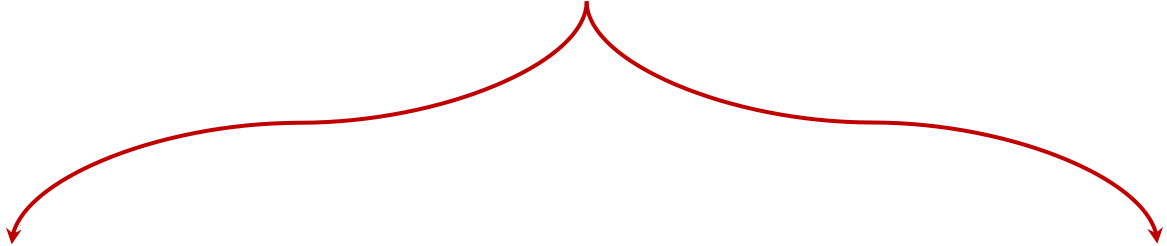
Outline

- Autonomous Systems
- From Manipulators to Mobile Robots
- **Mobile Robot Classifications**
- Challenging Problems of Mobile Robots
- Summary

Mobile Robot Classifications

Mobile robots may be classified based on the environment in which they travel.

Mobile Robots



```
graph TD; MR[Mobile Robots] --> IR[Indoor Robots]; MR --> OR[Outdoor Robots];
```

Indoor Robots

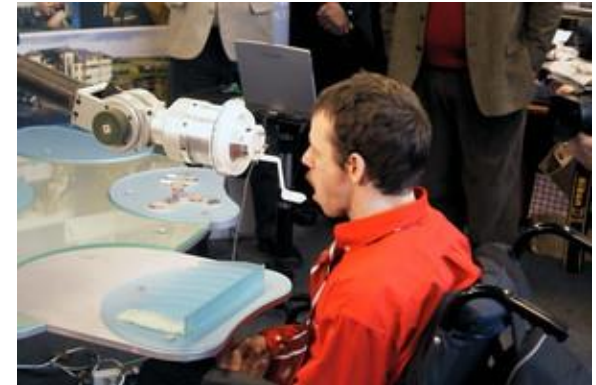
Outdoor Robots

Structured Environments

Unstructured Environments

Mobile Robot Classifications

- **Structured Environments**



Asibot

<http://roboticslab.uc3m.es/>



Carlos III University of Madrid



Mobile Robot Classifications

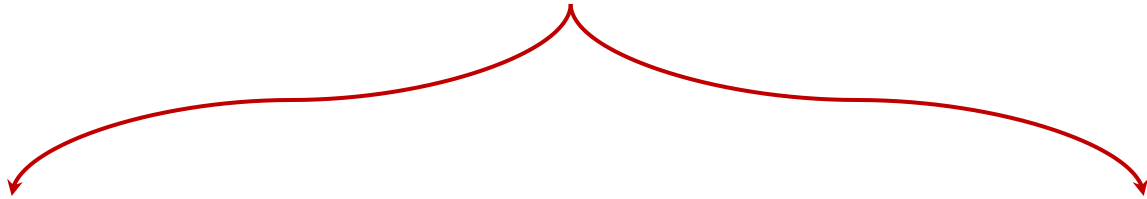
- Unstructured Environments



Boston Dynamics BigDog Robot
The Army mule

Mobile Robot Classifications

Mobile Robots



Indoor Robots

Structured Environments

- Research
- Transportation: industry & service
- Customer support, shops,...
- Cleaning
- Indoor Surveillance
- Assistive environments
- Entertainment, toy.

Outdoor Robots

Unstructured Environments

- Space exploration
- Mining, Demining
- Inspection
- Forest, Agriculture
- Construction
- Fire fighting
- Surveillance, Search & rescue
- Military, Spying.

Mobile Robot Classifications

- Indoor Robots: Research



B21



Koala



Magellan Pro



Hemisson



Khepera



Create

Mobile Robot Classifications

- **Indoor Robots: Transportation**

Automated Guided Vehicle (AGV) is a mobile robot that can be used to tow objects behind them in small trailers which they can autonomously hook up to.

These trailers can be used to move raw materials into line to get them ready to be manufactured. The AGV can also store objects on a bed. The objects can be placed on a set of motorized treads and then pushed off by reversing them. Some AGVs use fork lifts to lift objects for storage.

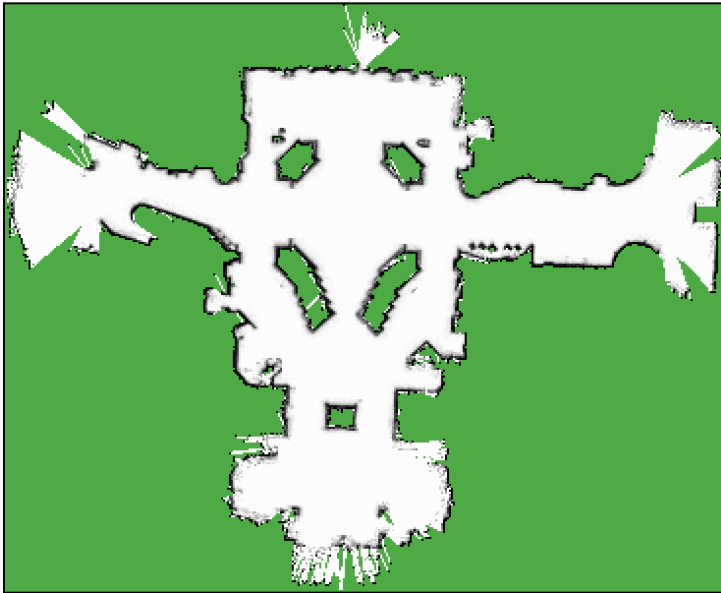
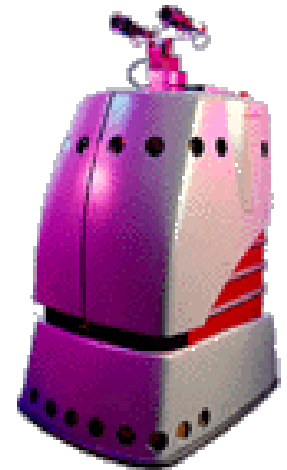
Transporting materials such as medicine in a hospital situation is also done.



Mobile Robot Classifications

- **Indoor Robots: Customer Support**

Minerva is an autonomous robot that moves daily through crowds at the Smithsonian's National Museum of American History.



Mobile Robot Classifications

- **Indoor Robots: Customer Support**

Unintelligent carts are commonly found in large airports. Travelers pick up carts at designated points and leave them in arbitrary places. It is a considerable task to re-collect them.



It is, therefore, desirable that intelligent carts (intelligent robots) draw themselves together automatically.



Mobile Robot Classifications

- **Indoor Robots: Cleaning**

Roomba - Making money with autonomous robots

Since the Roomba retails for about \$300 in 2003, it's been able to sell 1.5 million of them in three years.

iRobot®



Price now is about \$120



Mobile Robot Classifications

- **Indoor Robots: Cleaning**



Roomba - Making money with autonomous robots

A similar product called Scooba, which will clean and dry floors, is now available.



iRobot Scooba 5900
Floor Washing Robot
\$399.99



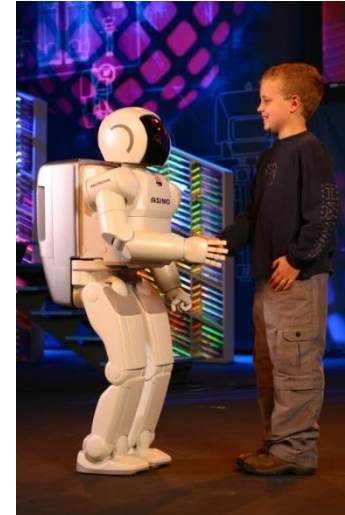
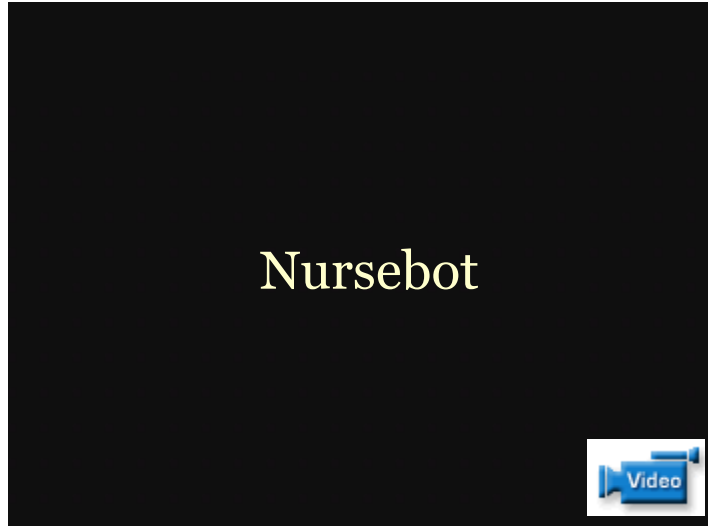
iRobot Scooba 5800
Floor Washing Robot
\$299.99

Mobile Robot Classifications

- Indoor Robots: Assistant Robots



Nursebot



ASIMO



Maggie
(UC3M)



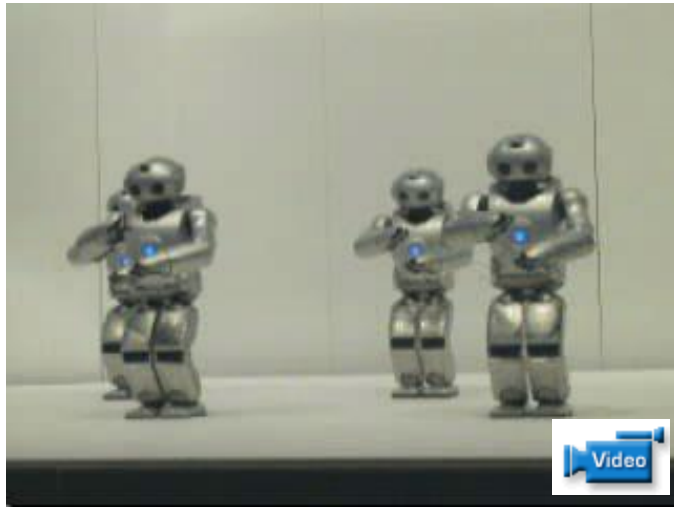
Wakamaru



RI-MAN

Mobile Robot Classifications

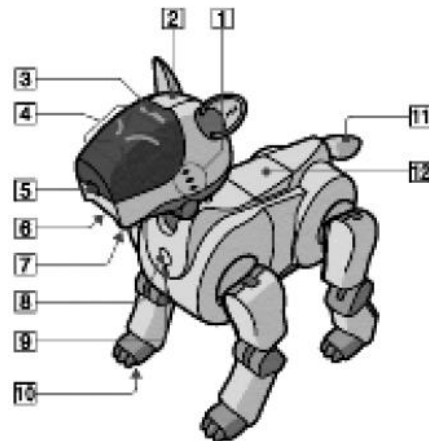
• Indoor Robots: Entertainment



AIBO



QRIO



- 1 Stereo microphone: Allows AIBO to pick up surrounding sounds.
- 2 Head sensor: Senses when a person taps or pets AIBO on the head.
- 3 Mode indicator: Shows AIBO's operation mode.
- 4 Eye lights: These light up in blue-green or red to indicate AIBO's emotional state.
- 5 Color camera: Allows AIBO to search for objects and recognize them by color and movement.
- 6 Speaker: Emits various musical tones and sound effects.
- 7 Chin sensor: Senses when a person touches AIBO on the chin.
- 8 Pause button: Press to activate AIBO or to pause AIBO.
- 9 Chest light: Gives information about the status of the robot.
- 10 Paw sensors: Located on the bottom of each paw.
- 11 Tail light: Lights up blue or orange to show AIBO's emotional state.
- 12 Back sensor: Senses when a person touches AIBO on the back.

Mobile Robot Classifications

- Indoor Robots: Entertainment



Anaconda

Mobile Robot Classifications

- **Outdoor Robots: Military**



“Future Combat System is a major program for an entire System of Systems to transform the U.S. Army to be strategically responsive and dominant at every point on the spectrum of operations, through real-time network-centric communications and systems for a family of manned vehicles and unmanned platforms by the next decade”.

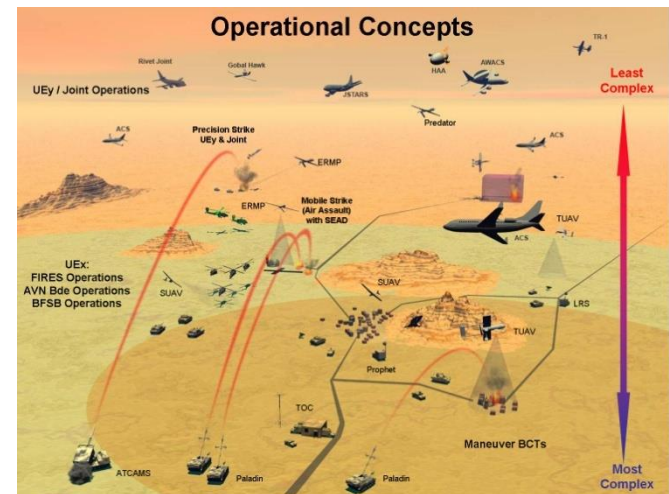


R-Gator: Autonomous Unmanned Ground Vehicle

Mobile Robot Classifications

- **Outdoor Robots: Military**

An unmanned aerial vehicle (UAV) is an aircraft with no onboard pilot. UAVs can be remote controlled aircraft (e.g. flown by a pilot at a ground control station) or can fly autonomously based on pre-programmed flight plans or more complex dynamic automation systems.



UAVs are currently used for a number of missions, including reconnaissance and attack roles.

Mobile Robot Classifications

- **Outdoor Robots: Military**

- There are 17,300 UAVs or drones in the US army inventory.
- These drones can carry up to 3000 pounds of weapons.
- Fabricated by Boeing



UAV carrying Viper
Strike Weapon System



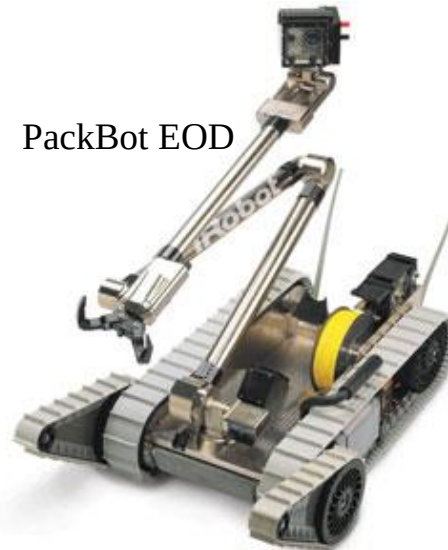
A forward looking infrared (FLIR)
camera mounted on the side of an UAV

Mobile Robot Classifications

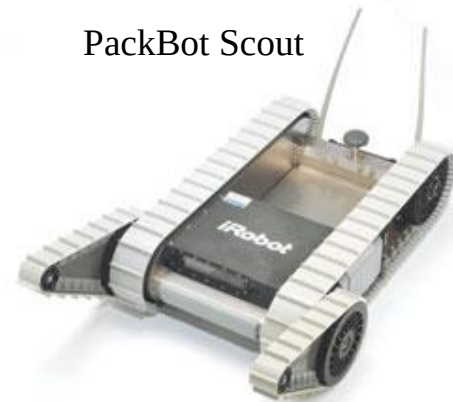
- **Outdoor Robots: Surveillance and Reconnaissance**



PackBot Explorer



PackBot EOD



PackBot Scout

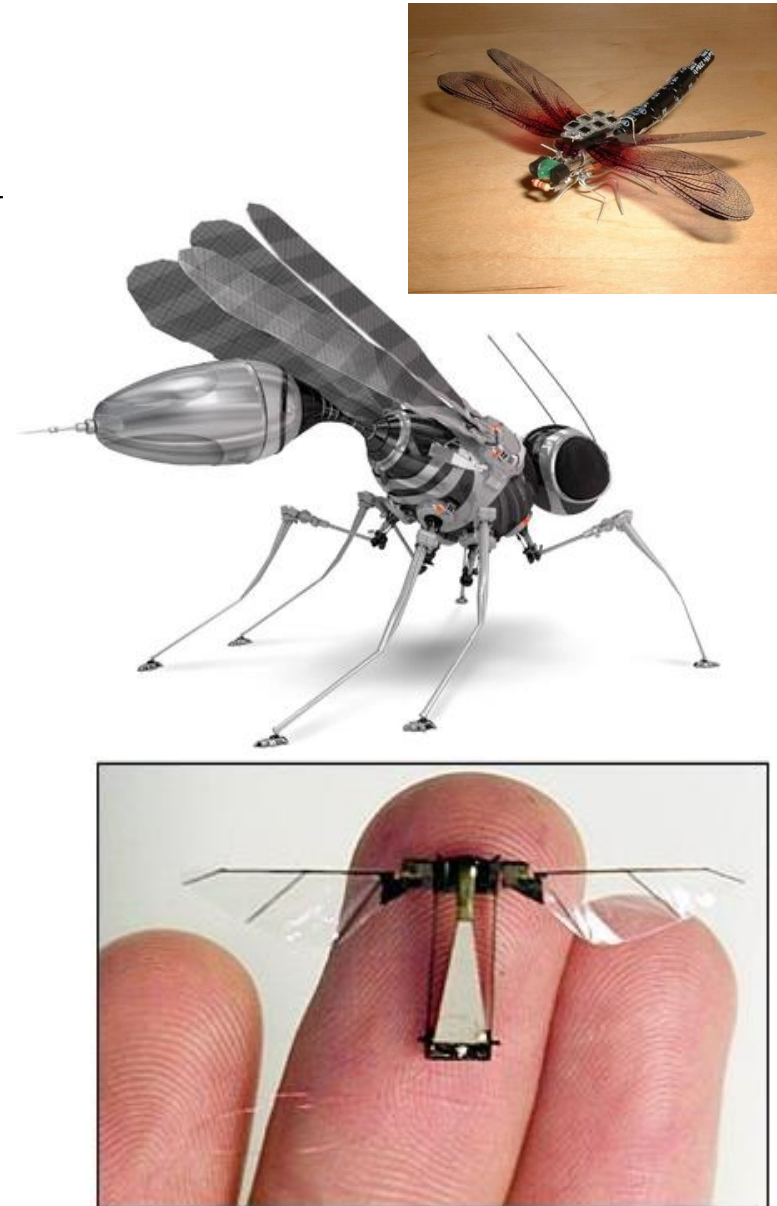
Mobile Robot Classifications

- **Outdoor Robots: Intelligence**

Micro Aerial Vehicles (MAV) are used in **spying missions**, where they quite literally serve as a “**fly on the wall**” - recording and transmitting audio-visual information.

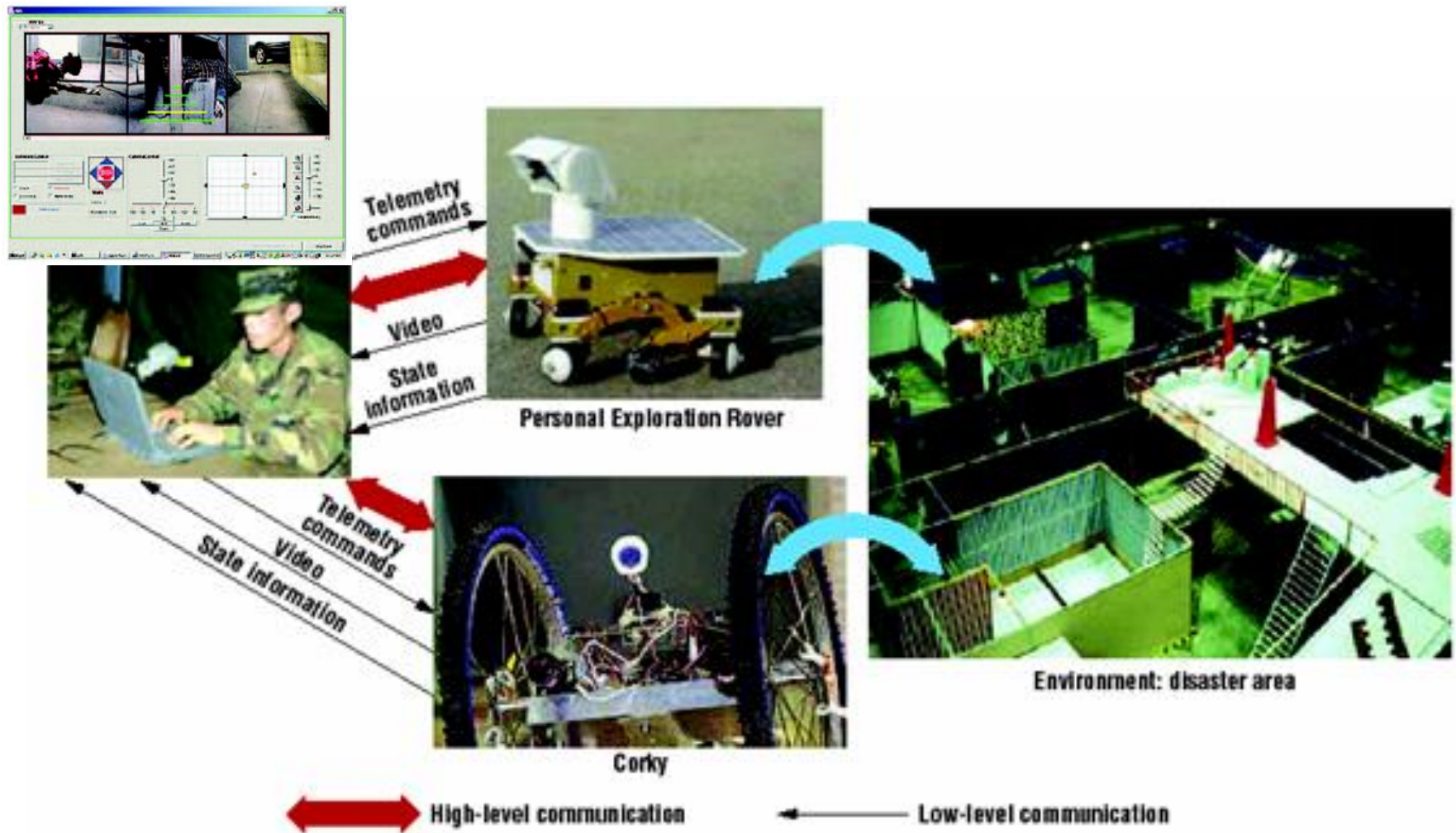
An individual robot is equipped with **miniature cameras, microphones, modem** and **GPS**.

A number of terrorist cells are being infiltrated thanks to this new technology.



Mobile Robot Classifications

- Outdoor Robots: Search and Rescue



Mobile Robot Classifications

- Outdoor Robots: Underwater



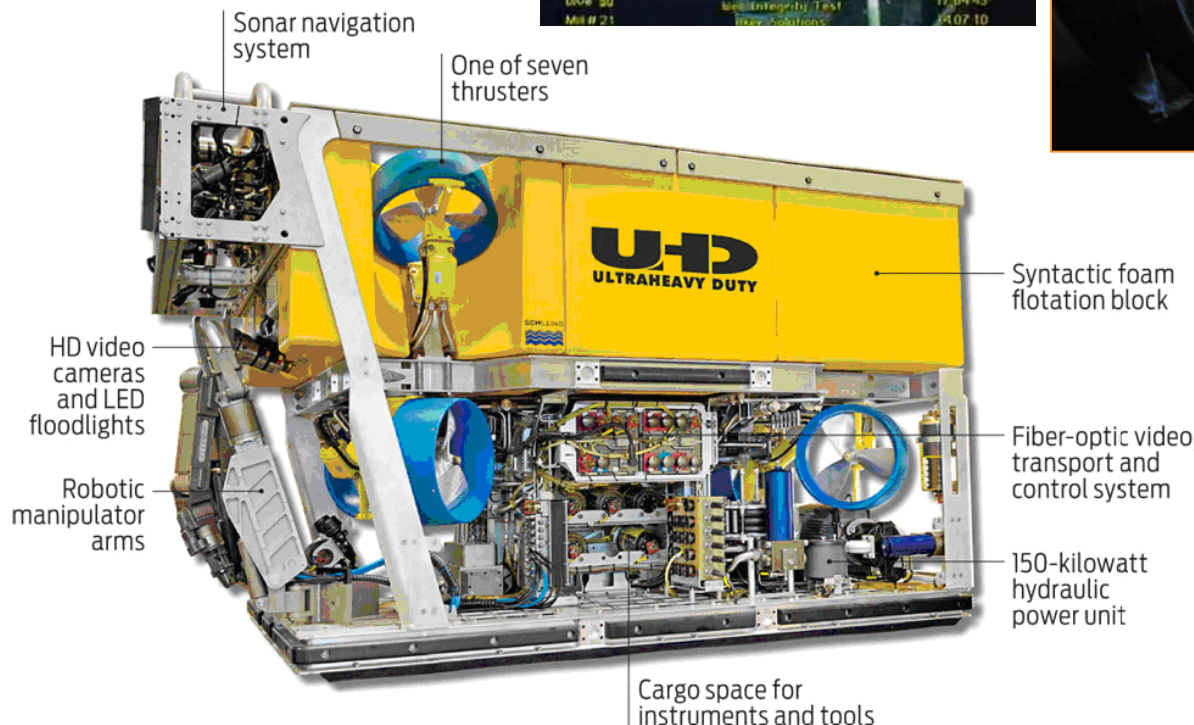
Mobile Robot Classifications

- Outdoor Robots: Underwater

BP's Deepwater Horizon oil rig problem in the Gulf of Mexico



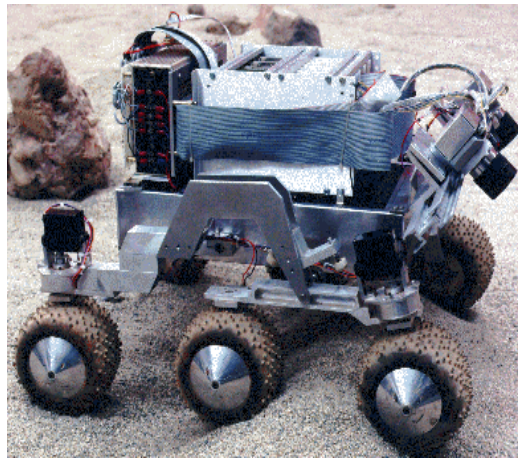
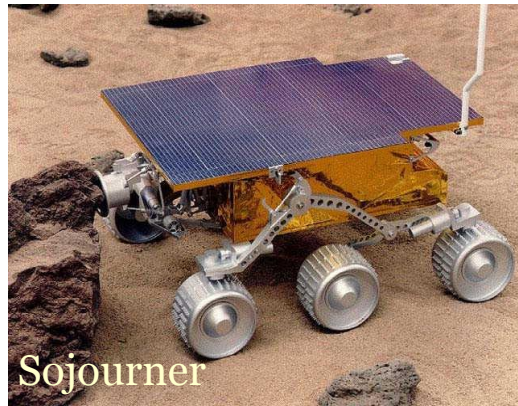
U.S. Coast Guard



Work-class ROVs are designed by Schilling Robotics to do power-intensive work hundreds of meters below the surface

Mobile Robot Classifications

- Outdoor Robots: Space Exploration Robots



Mobile Robot Classifications

• Outdoor Robots: Space Exploration Robots

NASA Mars mission

The Curiosity rover is designed to travel Mars studying climate and geology. The rover is looking for signs of carbon, the building blocks of life. Some of the rover's features:

Robotic arm

Used to examine and manipulate soil and rocks; it also has two scientific instruments, one uses X-rays to determine materials' composition and the other is a magnifying camera

Laser

Burns small holes in rocks and soil up to 23 feet away and identifies chemical elements

Color cameras

Stereo mastcams on either side of the rover's mast take color pictures and movies in 3-D

UHF antenna

Primary transmission antenna

Plutonium power source

A nuclear battery that converts heat into electricity

Neutron detector

Detects water in rocks and soil

Photo courtesy of NASA

Weather station

Records wind speed/direction, air pressure, humidity, temperature and UV radiation

Radiation detector

Measures radiation from the sun, supernovae and other sources

Inside:

Chemistry lab

Analyzes rock and soil samples for organics

Mineral detector

Shines an X-ray beam at a rock or soil sample to identify types of minerals



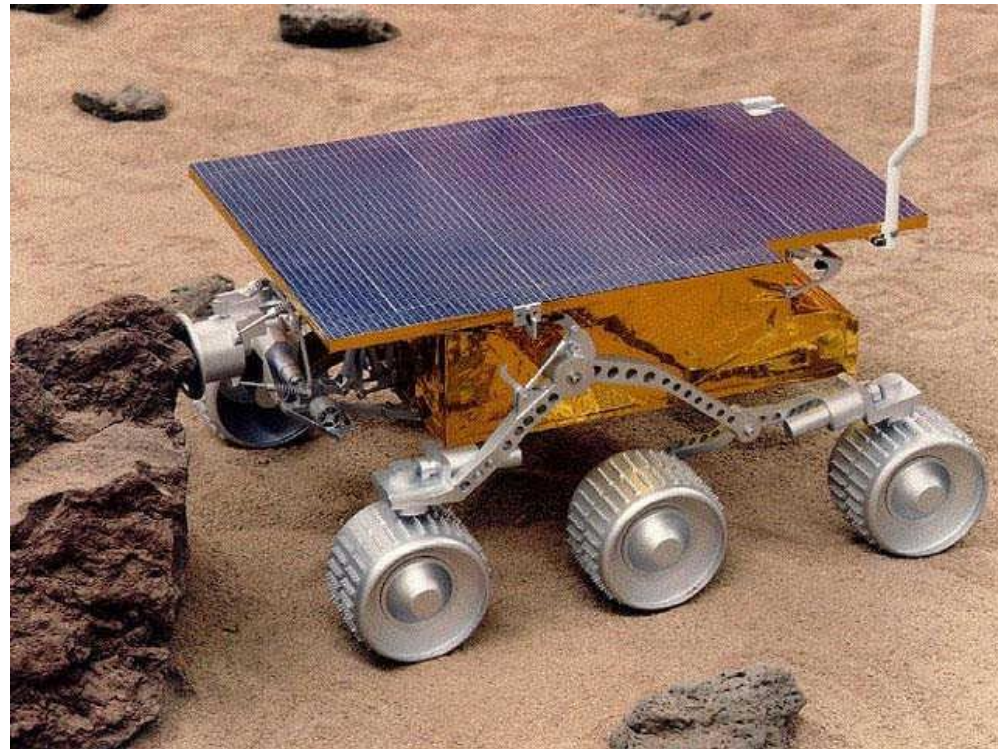
SOURCE: NASA

AP

Mobile Robot Classifications

- **Outdoor Robots: Space Exploration Robots**

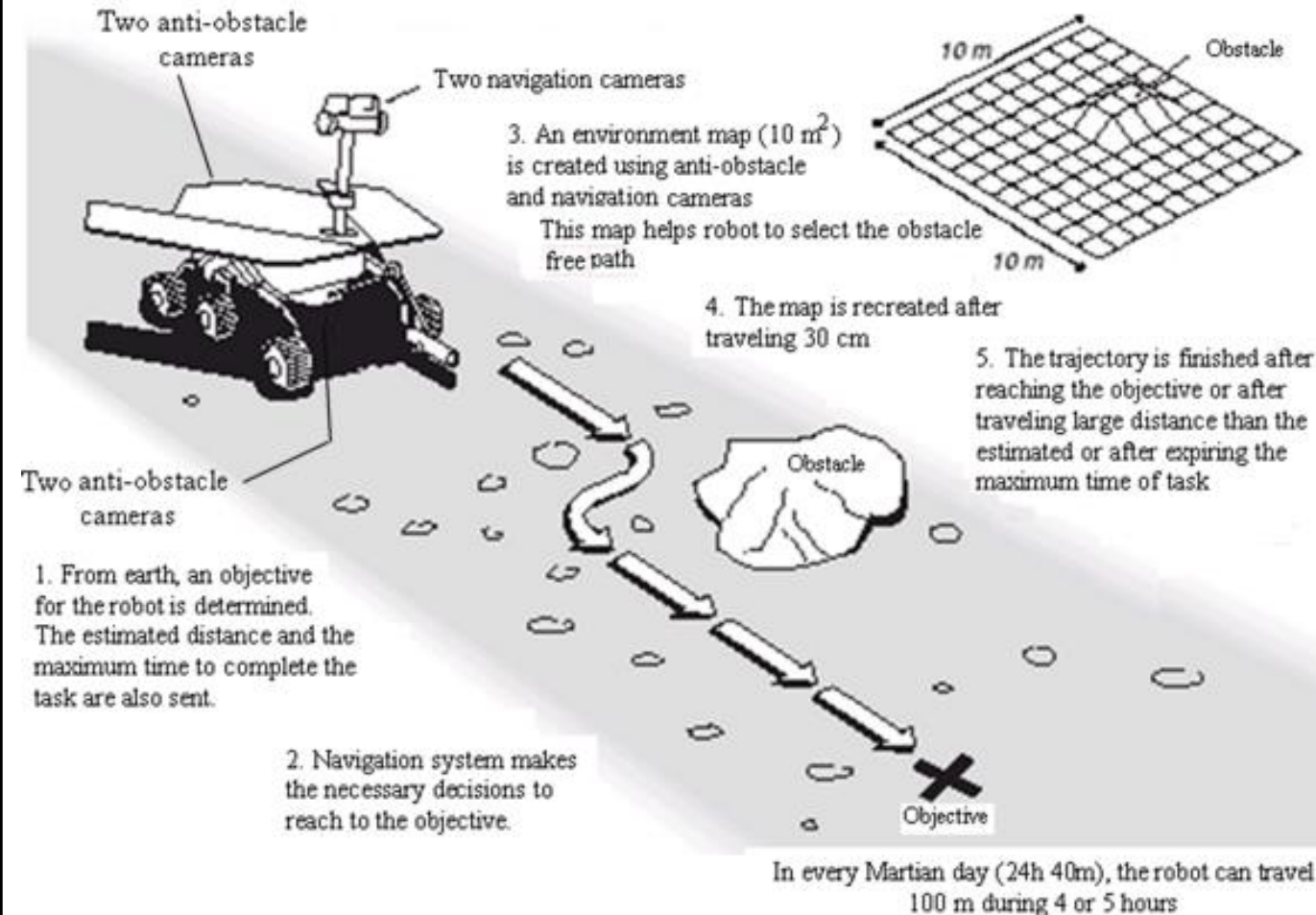
The mobile robot Sojourner was used during the Pathfinder mission to explore the mars in summer 1997. It was nearly fully teleoperated from earth. However, some on board sensors allowed for obstacle detection.



More info: http://ranier.oact.hq.nasa.gov/telerobotics_page/telerobotics.shtm

Mobile Robot Classifications

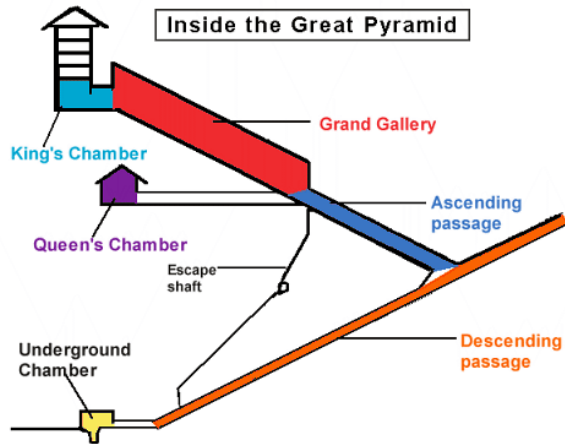
• Outdoor Robots: Space Exploration Robots



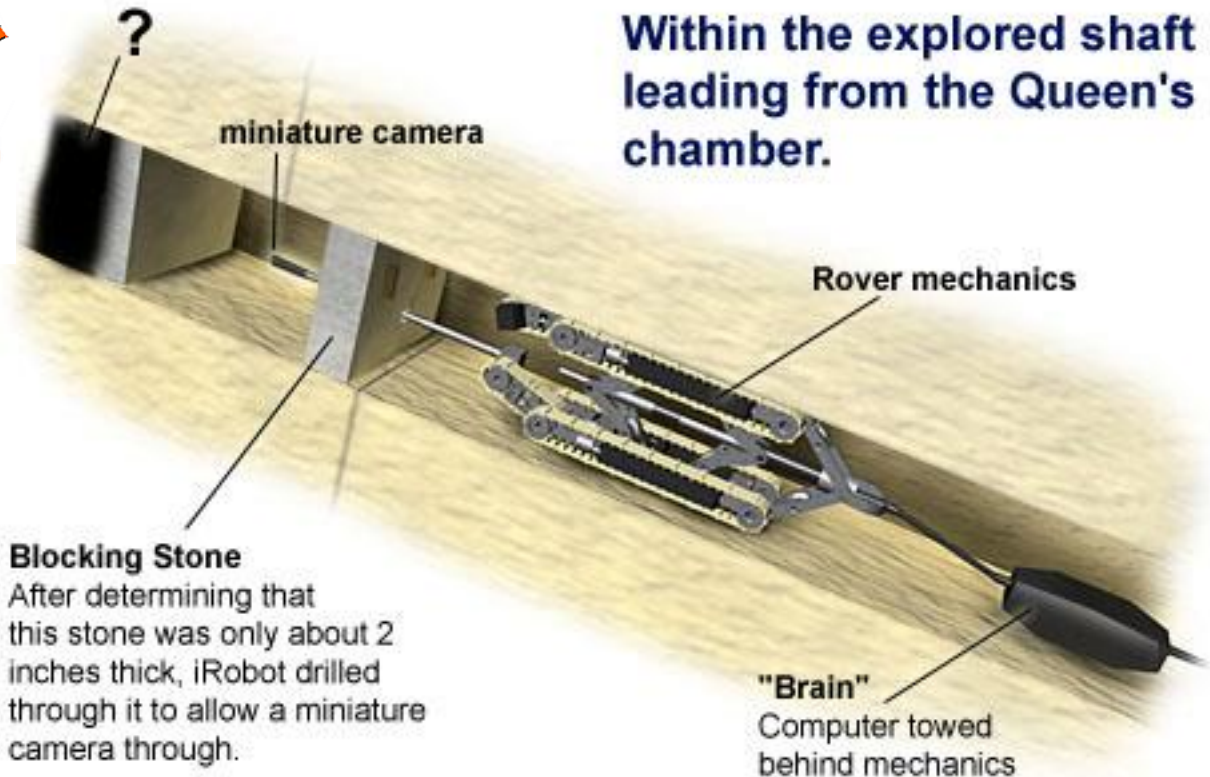
Mars Rover

Mobile Robot Classifications

- Outdoor Robots: Exploration



Within the explored shaft leading from the Queen's chamber.



Mobile Robot Classifications

- **Outdoor Robots: Fire Fighting**

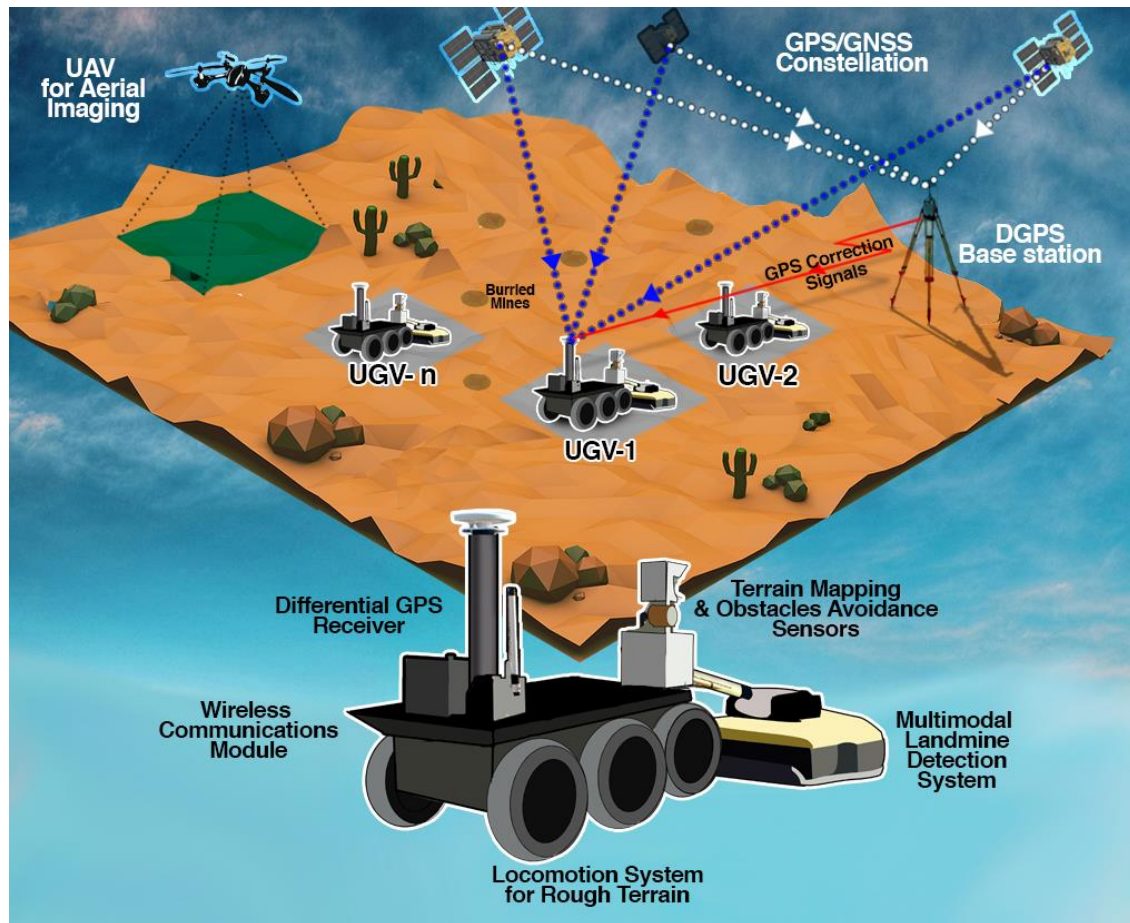


SACI 2.0 is a firefighting robot presented in July 2006 with the following features:

- It has Fault Tolerant Architecture with redundancy in its moving system;
- It has lighting system;
- It is modular;
- It has at least 100% more autonomy, with a full load capacity up to a minimum of six hours;
- It can have infrared or common video cameras and be controlled by wireless remote control.

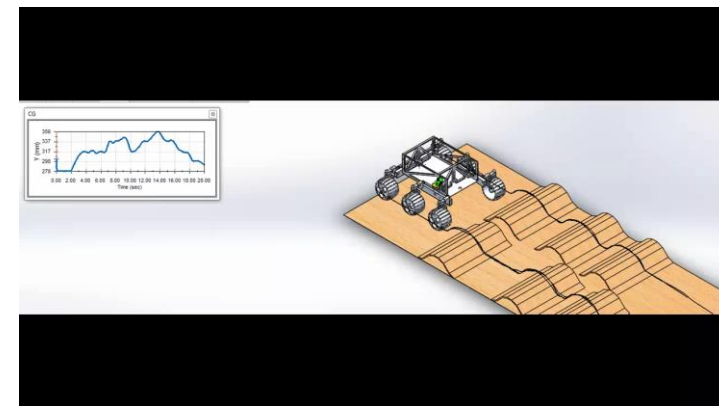
Mobile Robot Classifications

- Outdoor Robots: Demining – MineProbe



MineProbe: A Distributed Mobile Sensor System for Minefield Reconnaissance and Mapping in Egypt

<http://www.mineprobe.org>



Mobile Robot Classifications

- Outdoor Robots: Inspection

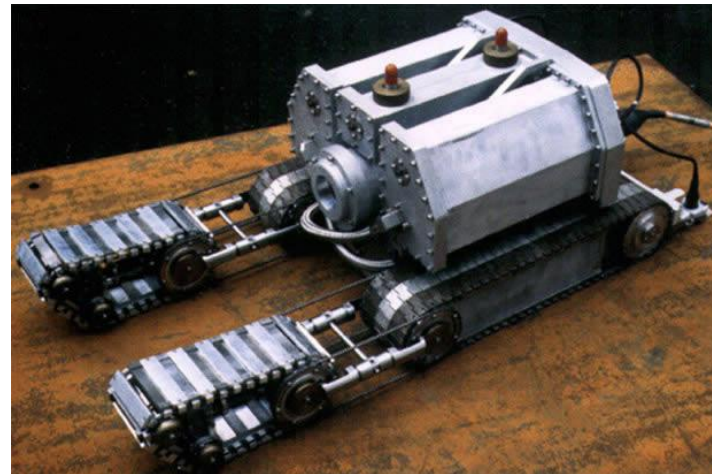


Pipeline Inspection



Pioneer

Neptune: A Robotic Inspection System for Oil Storage Tanks



Mobile Robot Classifications

- Outdoor Robots: Agriculture



Ag-robots Farm Robots



Demeter

National Robotics Engineering Consortium



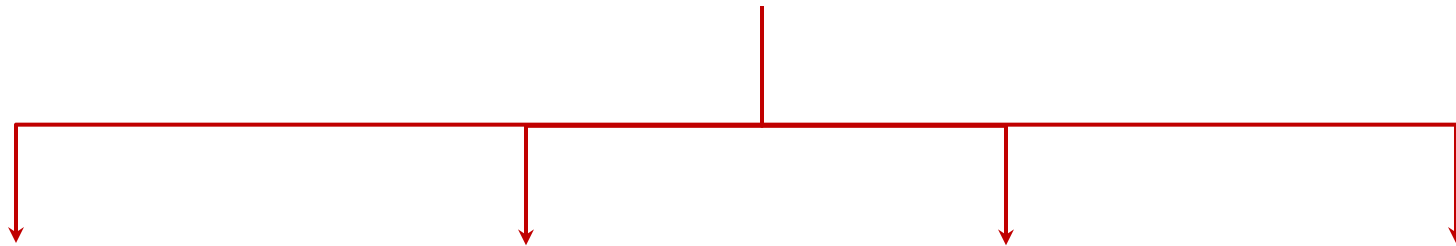
feldt-electronics : <http://www.feldt-electronics.de/> ASI Robotics: <https://www.asirobots.com/>



Mobile Robot Classifications

- Another classification : Unmanned Vehicles

UXVs



Unmanned Ground Vehicles (UGV)



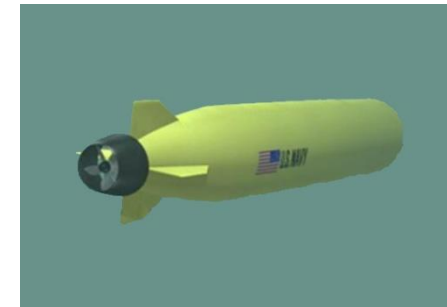
Unmanned Aerial Vehicles (UAV) & Micro Aerial Vehicles (MAV)



Unmanned Surface Vehicles (USV)



Unmanned Underwater Vehicles (UUV)



Outline

- Autonomous Systems
- From Manipulators to Mobile Robots
- Mobile Robot Classifications
- **Challenging Problems of Mobile Robots**
- Summary

Challenging Problems of Mobile Robots

- **Q1:** Where am I? Where am I going?
- **A1:** Localization
- **Q2:** What does the world look like?
- **A2:** Environment Mapping or Modeling
- **Q3:** How to explore an unknown environment?
- **A3:** Simultaneous Localization and Mapping “SLAM”
- **Q4:** How can I get there from here?
- **A4:** Planning
- **Q5:** How to achieve intelligent connection between perception and action?
- **A5:** Control Architecture



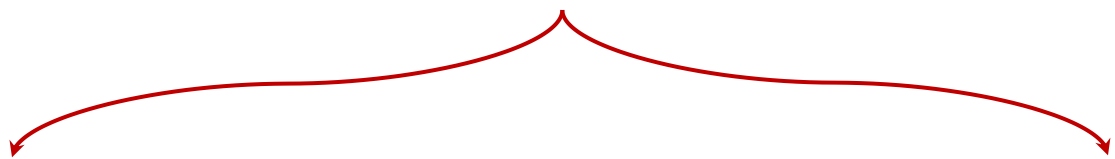
Challenging Problems of Mobile Robots

- **Q1: Where am I? Where am I going?**

A1: Localization

The localization process allows a mobile robot to know where it is at any moment relative to its environment.

Localization



Local Localization

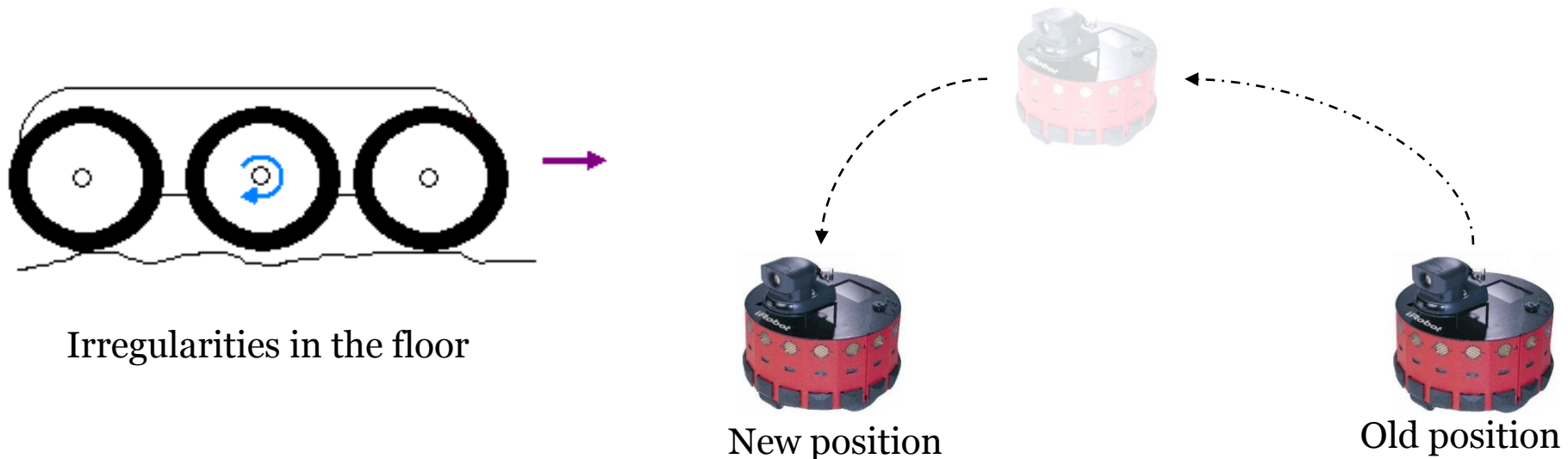
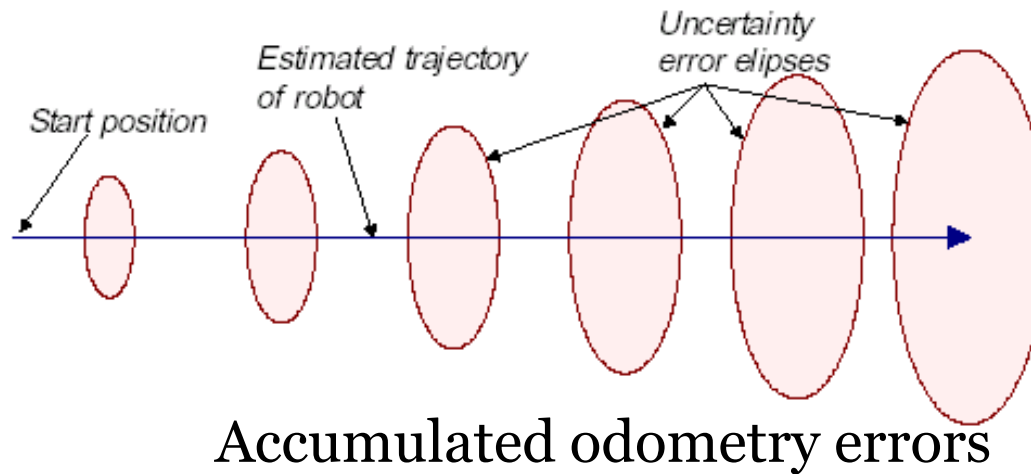
The robot incrementally corrects its position relative to an initial location

Global Localization

The robot's initial position is not needed. In addition, the location process can be based on the sensorial identification of **landmarks** in the environment whose location is well known, or it can be based on maps or models of the environment and identify characteristic elements of the mapped environment. Probabilistic approaches are used to solve the problem of uncertainty in the sensorial information.

Challenging Problems of Mobile Robots

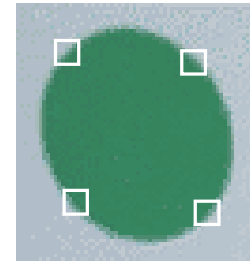
- Local Localization



Challenging Problems of Mobile Robots

- **Global Localization**

- Given a map, determine the robot's location
- Landmark locations are known, but the robot's position is not.



- From sensor readings, the robot must be able to infer its most likely position on the field.
- ***Example:*** where are the AIBOs on the soccer field?



Challenging Problems of Mobile Robots

- **Q2: What does the world look like?**

A2: Mapping

The information provided by the sensors has to be interpreted in a way that is sufficiently a structured knowledge of the space that surrounds the robot is obtained by the robot. Structuring this external information is accomplished through the construction of a **model or a map** of the environment.

This model is used to make decisions and to fuse the new information that the robot receives from its sensors, either from the **same sensor in several time instants** or from **several sensors** of the same or different type.



Challenging Problems of Mobile Robots

- Mapping

Maps

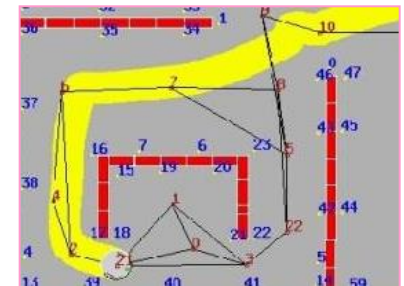
Metric Map

Metric maps capture the geometric properties of the environment.



Topological Map

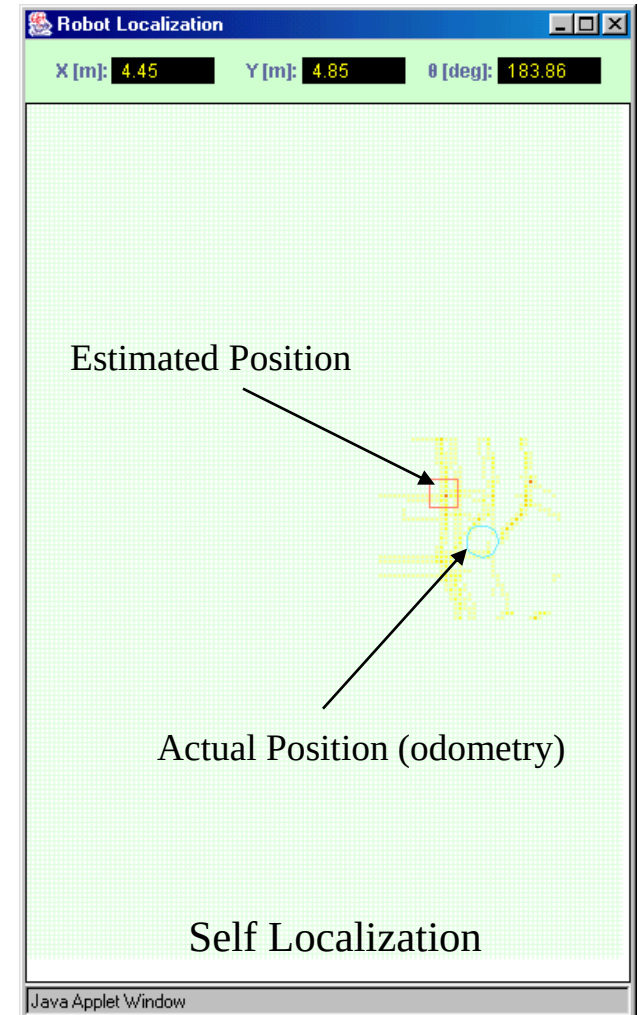
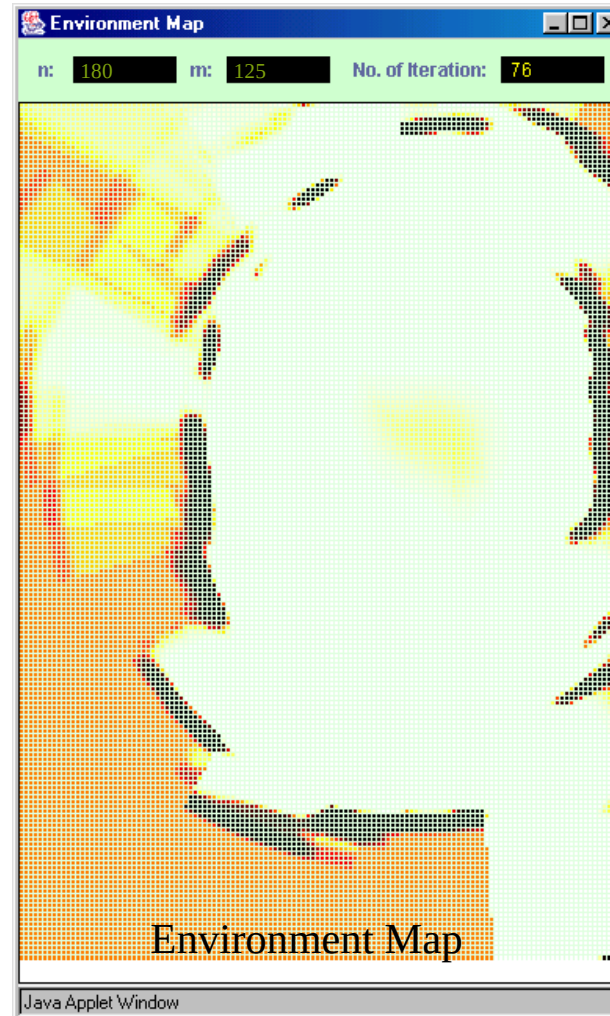
Topological maps describe the connectivity of different places by means of nodes-and-arcs graphs



Metric maps are finer grained than topological maps, but higher resolution comes at a computational burden. Metric maps can be discretized based on the probability of space occupation. The resulting mapping approaches are known as **occupancy-grid mapping**.

Challenging Problems of Mobile Robots

- Mapping and Localization



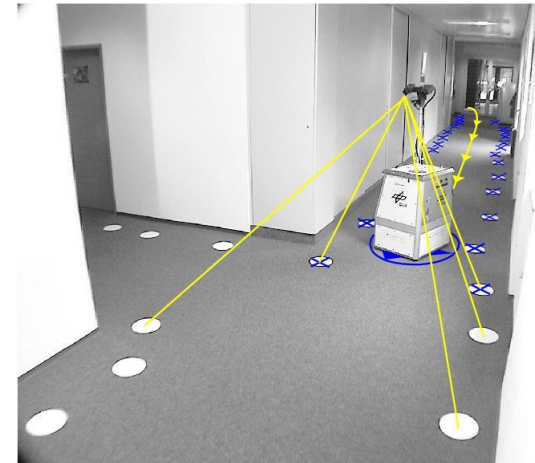
Challenging Problems of Mobile Robots

- **Q3: How to explore an unknown environment?**

A3: SLAM

Simultaneous localization and mapping (SLAM) is a technique used by robots and autonomous vehicles to build up a map within an unknown environment while at the same time keeping track of its current position.

This is not as straightforward as it might sound due to inherent uncertainties in discerning the robot's relative movement from its various sensors.



Challenging Problems of Mobile Robots

- **Q3: How to explore an unknown environment?**

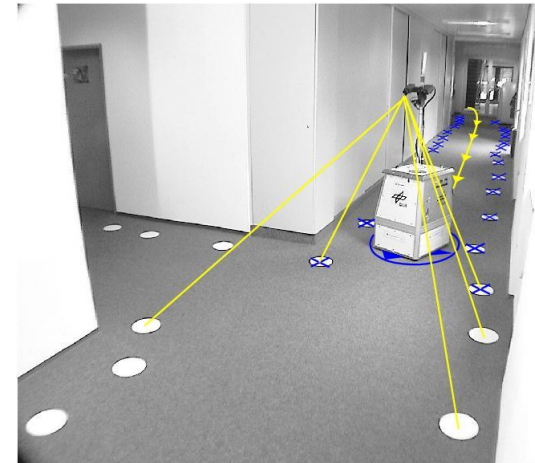
A3: SLAM

Inputs:

- No external coordinate reference.
- Time sequence of measurements made as robot moves through an initially unknown environment.

Outputs:

- A map of the environment.
- A robot pose estimate associated with each measurement, in the coordinate system in which the map is defined.



Challenging Problems of Mobile Robots

- Q4: How can I get there from here?

A4: Motion Planning

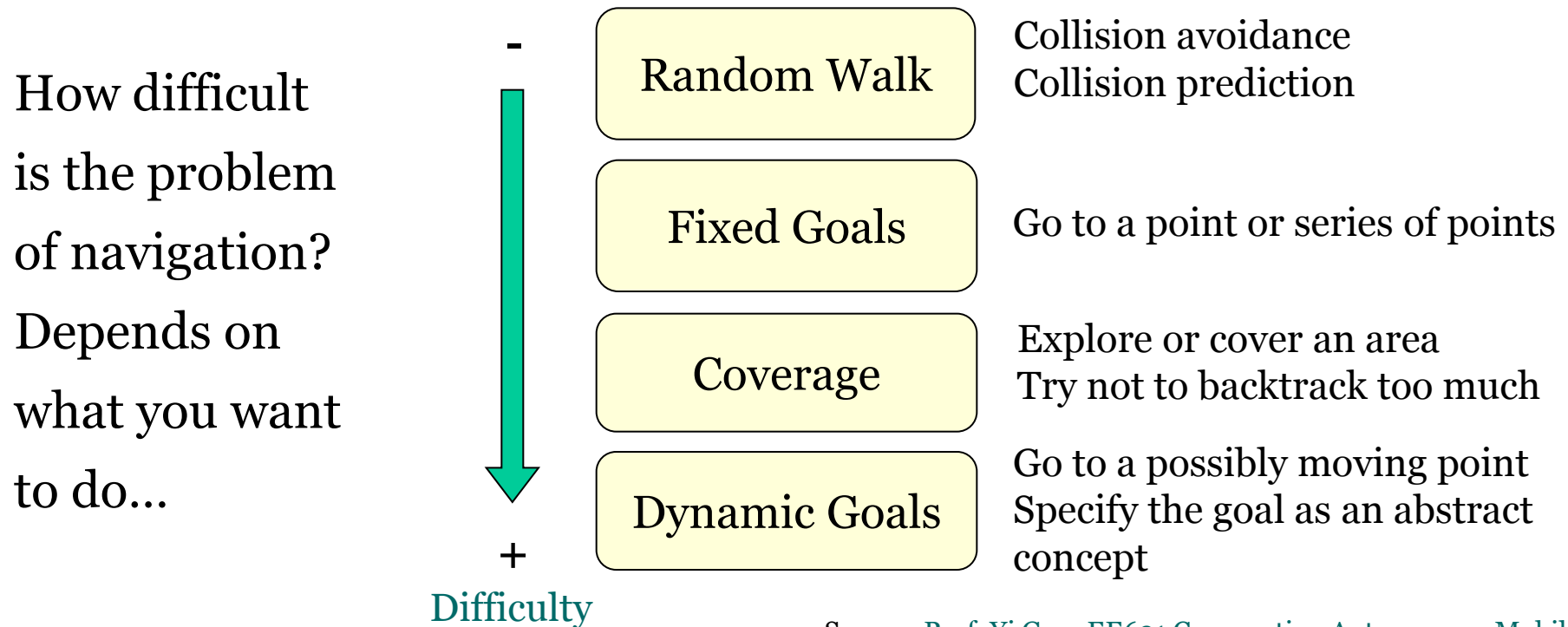


Path planning is a process to compute a **collision-free path** for a robot from a start position (s) to a given goal position (G), amidst a collection of obstacles.

Challenging Problems of Mobile Robots

- **Navigation**

Navigation is the action of **directing the course** of a mobile robot as it traverses the environment to make it reach a destination without getting lost or crashing with objects.



Source: Prof. Yi Guo, EE631 Cooperating Autonomous Mobile Robots, Stevens Institute of Technology, Hoboken, USA.

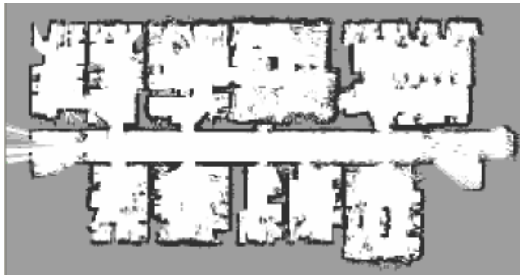
Challenging Problems of Mobile Robots

- Navigation

Navigation

Geometric Navigation

The robot pose, the environment elements and the goal are defined by **its coordinates**. The localization is calculated by **odometry**. The inevitable accumulated errors are reduced by measuring the relative position of the robot in relation to known elements of the environment elements. The environment is usually modeled by **occupancy grids** and by geometric maps which incorporate some elements of the environment (e.g. doors, walls, corners, artificial beacons, etc.). Occupancy grids are mainly used for path planning and geometric maps are used for relocalization.



Topological Navigation

Topologic navigation, also called **qualitative navigation**, is an alternative to geometric navigation. The information is established as relations, and usually represented with graphs. A main drawback of pure topological navigation is that robot localization is hard to manage and the robot can be lost quite easily.

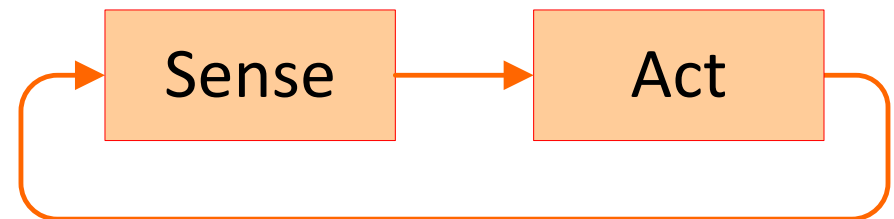


Challenging Problems of Mobile Robots

- **Navigation: Navigation Methods**

- 1. Reactive method: Sense-Decide-Act or Sense-Act**

- Simplify the sensing: use a lower level representation.
- Simplify the decision-making: use sub-symbolic or simple computation.
- Let the decision-making and analysis work in parallel.
- Combine the resulting “behaviors” in a useful way.
- Actions are much faster.

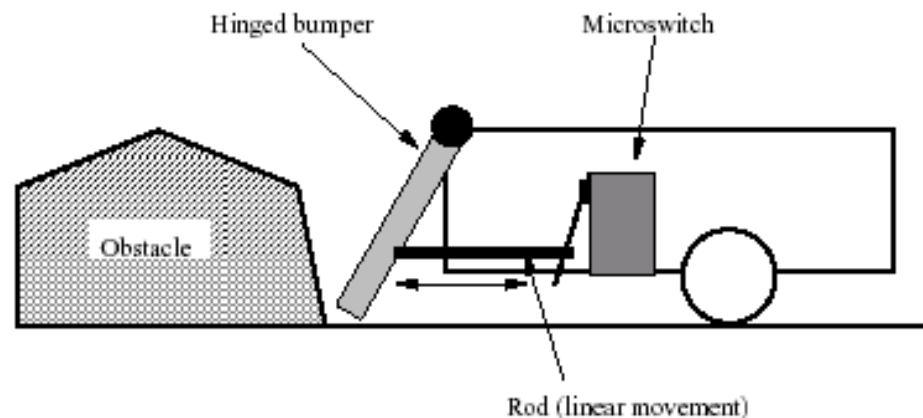


Challenging Problems of Mobile Robots

- **Navigation: Navigation Methods**

- 1. Reactive method: Sense-Decide-Act or Sense-Act**

Robot Primitives	Input	Output
Sense	Sensor data	Sensed information
Act	Sensed information	Actuator commands

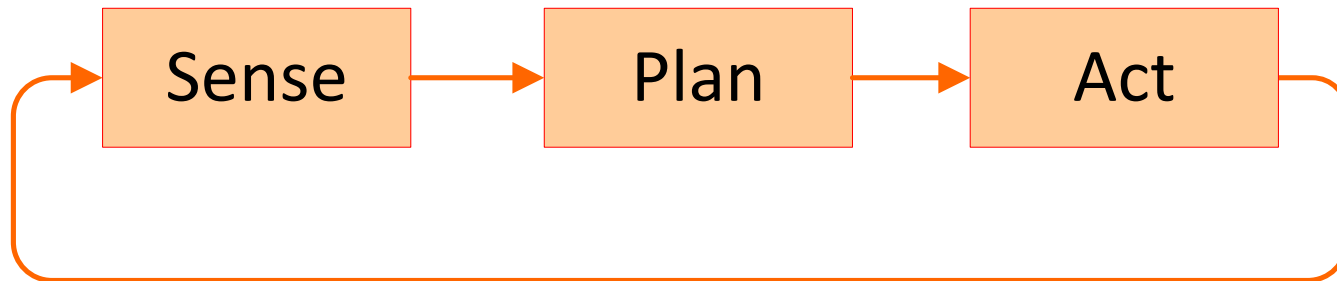


Challenging Problems of Mobile Robots

- **Navigation: Navigation Methods**

2. Deliberative method: Sense-Plan-Act

- Sensor analysis takes time: attempted to build high-level representations
- Planning is deliberative and takes time
- Actions are slow



Challenging Problems of Mobile Robots

- **Navigation: Navigation Methods**

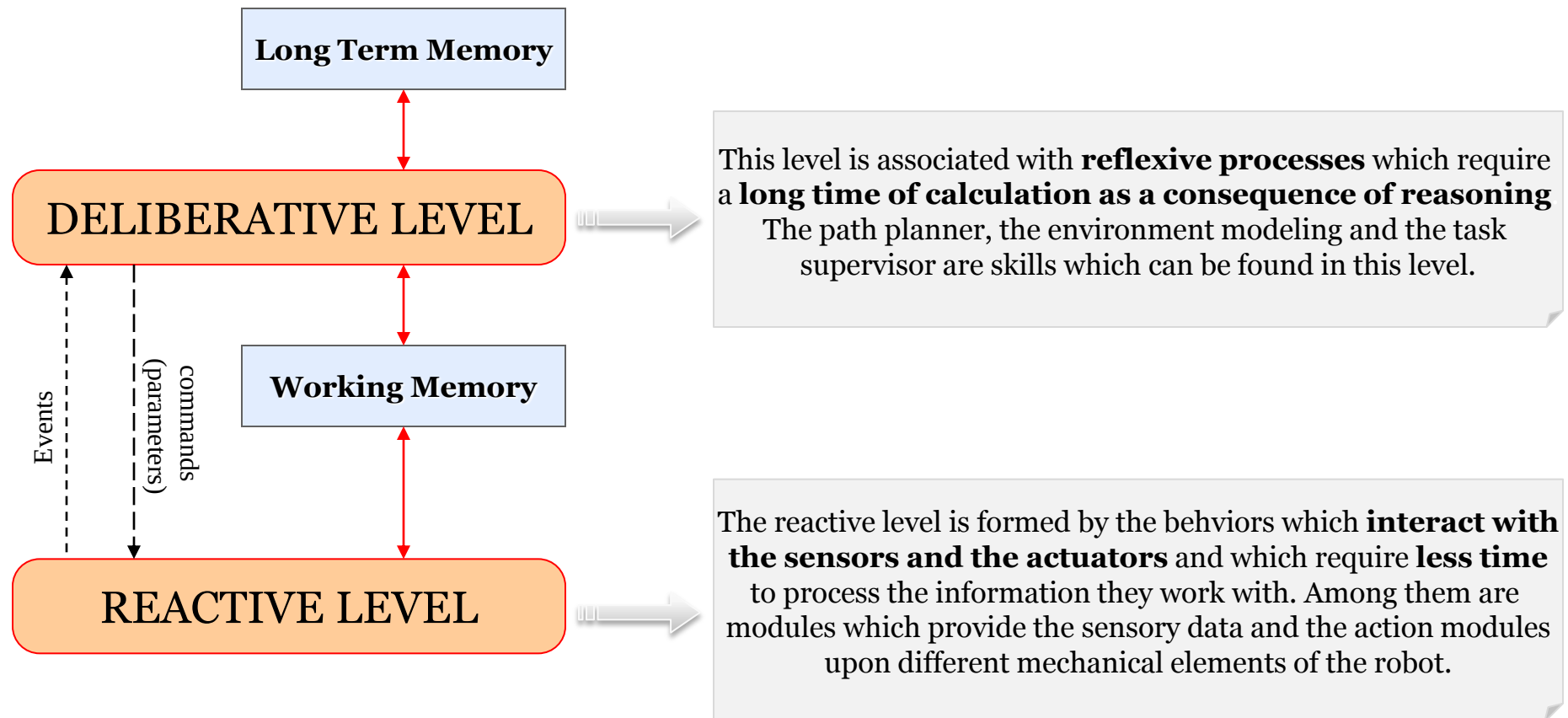
2. Deliberative method: Sense-Plan-Act

Robot Primitives	Input	Output
Sense	Sensor data	Sensed information
Plan	Information (sensed and/or cognitive)	Directives
Act	Sensed information or directives	Actuator commands

Challenging Problems of Mobile Robots

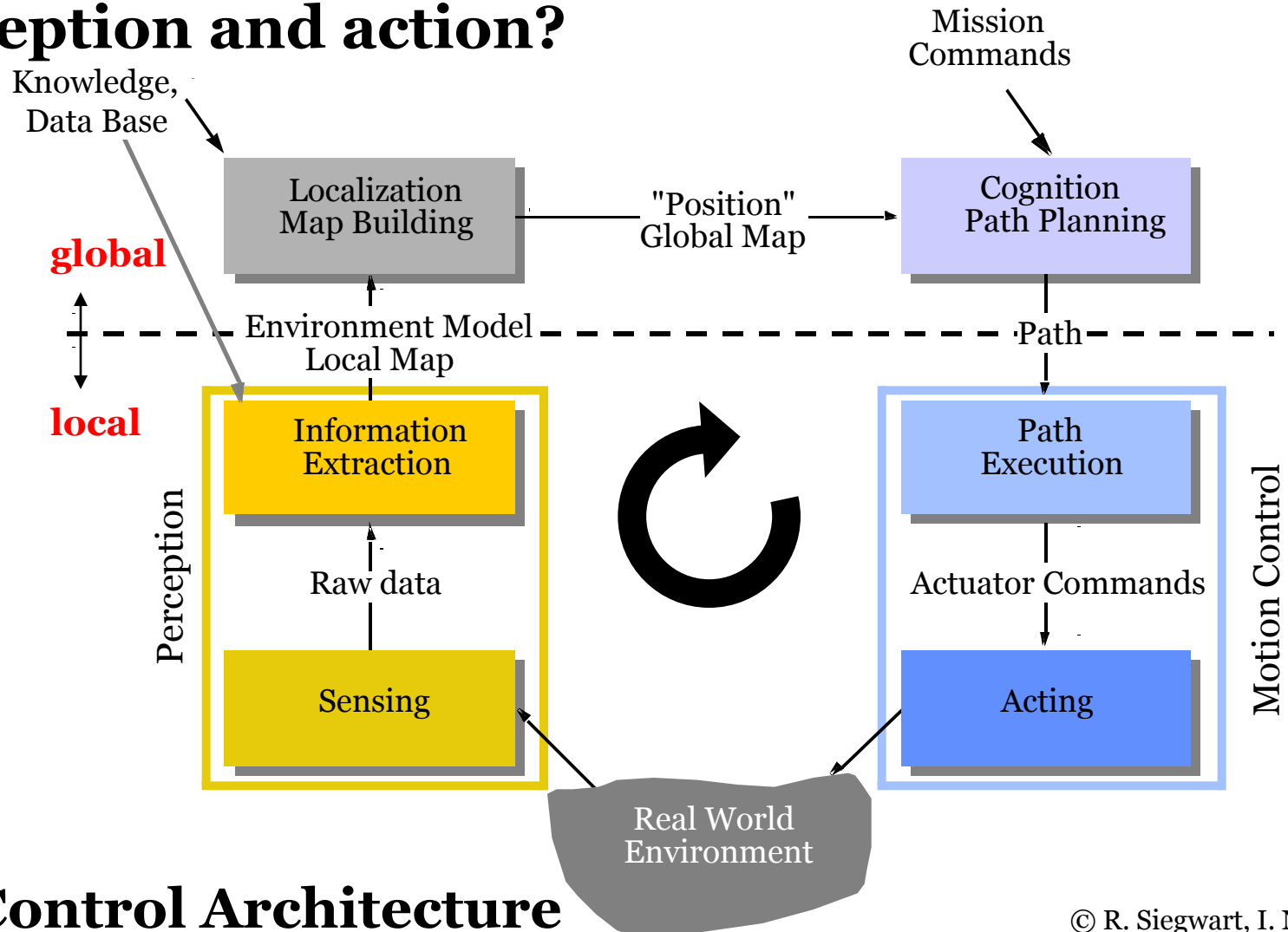
- **Navigation: Navigation Methods**

3. Combined Systems



Challenging Problems of Mobile Robots

- **Q5: How to achieve intelligent connection between perception and action?**



A5: Control Architecture

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Outline

- Autonomous Systems
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- **Summary**

Summary

- Mobile robotics is a young field. Its roots include many engineering and science disciplines, from mechanical, electrical and electronics engineering to computer, cognitive and social sciences.
- A mobile robot is a robot that can move in the real world and can be completely autonomous.
- Mobile robots include wheeled, tracked, legged, flying, and underwater robots.
- Mobile robots are commonly used as a platform to study different aspects of autonomous systems. These aspects include, but are not limited to, environment mapping, localization, motion planning, navigation, decision making under uncertainty, learning, and interaction with human or non-human actors in the environment.